

SOLIDWORKS API (.NET VBA C#) — Reference

(114/337): IFeatureManager (part 1 of 9)

SOLIDWORKS API Documentation (converted from CHM)

IFeatureManager (part 1 of 9) (Type)

SOLIDWORKS API Help

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IFeatureManager Interface

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[SolidWorks.Interop.sldworks Namespace](#) : IFeatureManager Interface

Allows you to create features.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface IFeatureManager
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager
```

C#

```
public interface IFeatureManager
```

C++/CLI

```
public interface class IFeatureManager
```

Visual Basic for Applications (VBA) Syntax

See [FeatureManager](#).

Example

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[IModelDoc2::FeatureManager](#)

Access Diagram

[FeatureManager](#)

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IFeatureManager Interface Methods

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[SolidWorks.Interop.sldworks Namespace](#) : IFeatureManager Interface

For a list of all members of this type, see [IFeatureManager members](#).

Public Methods

Name	Description
AddCornerReliefCorner	Adds the bend corner of two selected faces of a sheet metal body to the set of corners to which to apply a corner relief.
AddCornerReliefType	Specifies the type of corner relief to apply to the specified corner of the selected sheet metal body.
AddSMNormalCut	Obsolete. See IFeatureManager::CreateFeature and the Remarks in ISMNormalCutGroupData , and ISMNormalCutFeatureData2 .
AddSMNormalCutType	Obsolete. See IFeatureManager::CreateFeature and the Remarks in ISMNormalCutGroupData , and ISMNormalCutFeatureData2 .
AddVariablePitchHelixFirstPitchAndDiameter	Adds the first segment to a variable-pitch helix.
AddVariablePitchHelixSegment	Adds a segment to a variable-pitch helix.
AdvancedHole	Obsolete. Superseded by IFeatureManager::AdvancedHole2 .
AdvancedHole2	Creates an Advanced Hole feature.
CreateCoordinateSystem	Creates a coordinate system feature using the specified entities.
CreateCoordinateSystemUsingNumericalValues	Creates a coordinate system feature using the specified numerical values for position and orientation with respect to the global coordinate system.
CreateCustomBendAllowance	Creates a custom bend allowance to use when creating a sheet metal feature.
CreateDefinition	Creates a feature data object of the specified type.
CreateFeature	Creates the specified feature.
CreateFormTool	Obsolete. Superseded by IFeatureManager::CreateFormTool2 .
CreateFormTool2	Creates a forming tool feature with the specified point of insertion in a sheet metal part.
CreateSaveBodyFeature	Creates a Save Bodies feature and creates part and assembly documents of the save bodies.
CreateStructuralMemberGroup	Creates a weldment structural-member group.
CutListFolderLocation	Gets the parent folder of the specified solid body / cut list item / body folder.
DraftXpertChange	Changes the parameters on the selected drafted faces, regardless of whether the drafted faces were created manually or with DraftXpert, provided that DraftXpert can process them.
DraftXpertRemove	Deletes the draft on the selected faces. If all the faces of a draft are selected, then this method deletes the draft feature; if not, then this method edits the draft feature and removes the selected face references from it.
EditDeleteFace	Edits a DeleteFace feature.
EditFreeze	Obsolete. Superseded by IFeatureManager::EditFreeze2 .
EditFreeze2	Moves the freeze bar to the specified location in the FeatureManager design tree.
EditReferencePoint	Edits the selected reference points.
EditRollback	Rolls back or forward the rollback bar to a specific location in the FeatureManager design tree.
EndVariablePitchHelix	Ends and inserts a variable-pitch helix.
ExpandFeature	Expands the specified component in the specified FeatureManager design tree pane.
FeatureAdvancedTableDrivenPattern	Obsolete. See IFeatureManager::CreateFeature and the Remarks in IDimPatternFeatureData .
FeatureBossThicken	Thickens the selected reference surface, and then generates a boss.
FeatureChainPattern	Obsolete. See IFeatureManager::CreateFeature and the Remarks in IChainPatternFeatureData .
FeatureCircularPattern	Obsolete. Superseded by IFeatureManager::FeatureCircularPattern2 .
FeatureCircularPattern2	Obsolete. Superseded by IFeatureManager::FeatureCircularPattern3 .
FeatureCircularPattern3	Obsolete. Superseded by IFeatureManager::FeatureCircularPattern4 .
FeatureCircularPattern4	Obsolete. Superseded by IFeatureManager::FeatureCircularPattern5 .
FeatureCircularPattern5	Obsolete. See IFeatureManager::CreateFeature and the Remarks in ICircularPatternFeatureData and ILocalCircularPatternFeatureData .

	FeatureCut	Obsolete. Superseded by IFeatureManager.FeatureCut2 .
	FeatureCut2	Obsolete. Superseded by IFeatureManager.FeatureCut3 .
	FeatureCut3	Obsolete. Superseded by IFeatureManager.FeatureCut4 .
	FeatureCut4	Creates a cut extrude feature.
	FeatureCutThicken	Thickens the selected reference surface feature, and then generates a cut.
	FeatureCutThin	Obsolete. Superseded by IFeatureManager.FeatureCutThin2 .
	FeatureCutThin2	Inserts a thin cut extrude feature.
	FeatureDimensionPattern	Not implemented.
	FeatureExtruRefSurface	Obsolete. Superseded by IFeatureManager.FeatureExtruRefSurface2 .
	FeatureExtruRefSurface2	Obsolete. Superseded by IFeatureManager.FeatureExtruRefSurface3 .
	FeatureExtruRefSurface3	Inserts an extruded surface.
	FeatureExtrusion	Obsolete. Superseded by IFeatureManager.FeatureExtrusion2 .
	FeatureExtrusion2	Obsolete. Superseded by IFeatureManager.FeatureExtrusion3 .
	FeatureExtrusion3	Creates an extruded feature.
	FeatureExtrusionThin	Obsolete. Superseded by IFeatureManager.FeatureExtrusionThin2 .
	FeatureExtrusionThin2	Creates an extruded thin feature.
	FeatureFillet	Obsolete. Superseded by IFeatureManager.FeatureFillet2 .
	FeatureFillet2	Obsolete. Superseded by IFeatureManager.FeatureFillet3 .
	FeatureFillet3	Creates a fillet feature for selected edges and control point references.
	FeatureFillPattern	Obsolete. See IFeatureManager.CreateFeature and the Remarks in IFillPatternFeatureData .
	FeatureFolderLocation	Gets the folder feature for the specified feature.
	FeatureLinearPattern	Obsolete. Superseded by IFeatureManager.FeatureLinearPattern2 .
	FeatureLinearPattern2	Obsolete. Superseded by IFeatureManager.FeatureLinearPattern3 .
	FeatureLinearPattern3	Obsolete. Superseded by IFeatureManager.FeatureLinearPattern4 .
	FeatureLinearPattern4	Obsolete. Superseded by IFeatureManager.FeatureLinearPattern5 .
	FeatureLinearPattern5	Obsolete. See IFeatureManager.CreateFeature and the Remarks in ILinearPatternFeatureData and ILocalLinearPatternFeatureData .
	FeatureLocalCurveDrivenPattern	Obsolete. See IFeatureManager.CreateFeature and the Remarks in ILocalCurvePatternFeatureData .
	FeatureLocalSketchDrivenPattern	Obsolete. See IFeatureManager.CreateFeature and the Remarks in ILocalSketchPatternFeatureData .
	FeatureRevolve	Obsolete. Superseded by IFeatureManager.FeatureRevolve2 .
	FeatureRevolve2	Creates a base-, boss-, or cut-revolve feature.
	FeatureRevolveCut	Obsolete. Superseded by IFeatureManager.FeatureRevolveCut2 .
	FeatureRevolveCut2	Obsolete. Superseded by IFeatureManager.FeatureRevolve2 .
	FeatureRevolveThin	Obsolete. Superseded by IFeatureManager.FeatureRevolve2 .
	FeatureRevolveThinCut	Obsolete. Superseded by IFeatureManager.FeatureRevolve2 .
	FeatureSketchDrivenPattern	Obsolete. See IFeatureManager.CreateFeature and the Remarks in ISketchPatternFeatureData .
	FilletXpertChange	Changes the parameters on the selected filleted faces, regardless of whether the filleted faces were created manually or with FilletXpert, provided that FilletXpert can process them.
	FilletXpertMakeCorner	Creates or changes a fillet corner feature.
	FilletXpertRemove	Deletes the fillets on the selected faces.
	FinishCornerRelief	Creates a sheet metal corner relief feature.
	FinishSMNormalCut	Obsolete. See IFeatureManager.CreateFeature and the Remarks in ISMNormalCutGroupData , and ISMNormalCutFeatureData2 .
	GetCornerManagementFolders	Gets the structure system corner management folders.
	GetCreateFeatureErrors	Gets the errors that occurred during the last call to IFeatureManager.CreateFeature .
	GetFeatureCount	Gets the number of features in this document.

		
	GetFeatures	Gets the features in this document.
	GetFeatureTreeRootItem	Obsolete. Superseded by IFeatureManager::GetFeatureTreeRootItem2 .
	GetFeatureTreeRootItem2	Gets the root item of the FeatureManager design tree in the specified pane.
	GetFlatPatternFolder	Gets the interface to the flat-pattern folder feature in the FeatureManager design tree.
	GetFreezeLocation	Gets the location of the freeze bar in the FeatureManager design tree.
	GetPreTrimmedBodies	Gets the temporary trimmed bodies using the specified target sheet (surface) body according to the trim tools previously defined by IFeatureManager::PreTrimSurface .
	GetSelectionSetFolder	Gets the Selection Sets folder.
	GetSheetMetalFolder	Gets the interface to the sheet metal folder feature in the FeatureManager design tree.
	GetSlicingData	Gets a new slicing data object with default parameter values.
	GetStructureSystemFolders	Gets the structure system folders.
	HideBodies	Hides both solid and surface bodies in the model.
	HoleWizard	Obsolete. Superseded by IFeatureManager::HoleWizard2 .
	HoleWizard2	Obsolete. Superseded by IFeatureManager::HoleWizard3 .
	HoleWizard3	Obsolete. Superseded by IFeatureManager::HoleWizard4 .
	HoleWizard4	Obsolete. Superseded by IFeatureManager::HoleWizard5 .
	HoleWizard5	Creates customized holes of various kinds.
	IFeatureFillet	Obsolete. Superseded by IFeatureManager::IFeatureFillet2 .
	IFeatureFillet2	Obsolete. Superseded by IFeatureManager::FeatureFillet3 .
	IGetFeatures	Gets the features in this document.
	InsertCombineFeature	Combines the specified bodies in the multibody part to create a combine feature.
	InsertMacroFeature	Obsolete. Superseded by IFeatureManager::InsertMacroFeature3 .
	InsertMacroFeature2	Obsolete. Superseded by IFeatureManager::InsertMacroFeature3 .
	InsertMacroFeature3	Inserts a macro feature in this model.
	InsertMoveFace2	Obsolete. Superseded by IFeatureManager::InsertMoveFace3 .
	InsertReferencePoint	Creates the geometry for the reference points based on any of these selected entities: edges, faces, planes, vertices, or sketch geometry.
	InsertSheetMetalEdgeFlange2	Obsolete. Superseded by IFeatureManager::CreateDefinition and IFeatureManager::CreateFeature .
	InsertTableDrivenPattern	Obsolete. Superseded by IFeatureManager::InsertTableDrivenPattern2 .
	InsertCenterOfMass	Inserts a Center of Mass feature.
	InsertCenterOfMassReferencePoint	Inserts a Center of Mass Reference Point feature.
	InsertCombineFeature	Combines the specified bodies in the multibody part to create a combine feature.
	InsertConnectionPoint	Adds a connection point based on the selected entities.
	InsertConvertToSheetMetal	Obsolete. Superseded by IFeatureManager::InsertConvertToSheetMetal2 .
	InsertConvertToSheetMetal2	Obsolete. Superseded by IFeatureManager::CreateDefinition and IFeatureManager::CreateFeature .
	InsertCoordinateSystem	Inserts a coordinate system feature.
	InsertCosmeticThread	Obsolete. Superseded by IFeatureManager::InsertCosmeticThread2 .
	InsertCosmeticThread2	Obsolete. Superseded by IFeatureManager::InsertCosmeticThread3 .
	InsertCosmeticThread3	Inserts a cosmetic thread.
	InsertCosmeticWeldBead	Obsolete. Superseded by IFeatureManager::InsertCosmeticWeldBead2 .
	InsertCosmeticWeldBead2	Inserts a cosmetic weld bead using either weld geometry or a weld path.
	InsertCrossBreak	Inserts a cross break feature on the selected face in a sheet metal part.
	InsertCutBlend	Inserts a lofted cut based on the selected profiles, centerline, and guide curves.
	InsertCutSurface	Inserts a surface-cut feature using the preselected surface or plane.

		
	InsertCutSwept	Obsolete. Superseded by IFeatureManager.InsertCutSwept3 .
	InsertCutSwept2	Obsolete. Superseded by IFeatureManager.InsertCutSwept3 .
	InsertCutSwept3	Obsolete. Superseded by IFeatureManager.InsertCutSwept4 .
	InsertCutSwept4	Obsolete. Superseded by IFeatureManager.InsertCutSwept5 .
	InsertCutSwept5	Obsolete. See Remarks .
	InsertDeleteBody	Obsolete. Superseded by IFeatureManager.InsertDeleteBody2 .
	InsertDeleteBody2	Inserts a Body-Delete/Keep feature.
	InsertDeleteHoleForSurface	Inserts a Delete Hole feature for one or more selected hole edges on a surface.
	InsertDerivedPattern	Obsolete. Superseded by IFeatureManager.InsertDerivedPattern2 .
	InsertDerivedPattern2	Obsolete. See IFeatureManager.CreateFeature and the Remarks in IDerivedPatternFeatureData .
	InsertDwgOrDxfFile	Obsolete. Superseded by IFeatureManager.InsertDwgOrDxfFile2 .
	InsertDwgOrDxfFile2	Inserts a DXF/DWG image feature.
	InsertEdgeMerge	Merges multiple edges into a single edge using the selected faces when importing data.
	InsertEndCapFeature	Obsolete. Superseded by IFeatureManager.InsertEndCapFeature3 .
	InsertEndCapFeature2	Inserts an end cap feature using the specified end faces of a structural member.
	InsertEndCapFeature3	Inserts an end cap feature for one or more pre-selected open ends of a structural member.
	InsertFeatureChamfer	Inserts a chamfer.
	InsertFeatureLock	Locks or freezes a selected feature.
	InsertFeatureTreeFolder	Obsolete. Superseded by IFeatureManager.InsertFeatureTreeFolder2 .
	InsertFeatureTreeFolder2	Inserts a folder in the FeatureManager design tree for pre-selected features or components.
	InsertFilletBeadFeature	Obsolete. Superseded by IFeatureManager.InsertFilletBeadFeature2 .
	InsertFilletBeadFeature2	Inserts a fillet weld bead feature and also fills the gap between the pre-selected part bodies, if any.
	InsertFilletBeadFeature3	Inserts fillet weld bead features for the specified face sets.
	InsertFillSurface	Obsolete. Superseded by IFeatureManager.InsertFillSurface2 .
	InsertFillSurface2	Inserts a fill-surface feature in the model.
	InsertFlattenSurface	Obsolete. Superseded by IFeatureManager.InsertFlattenSurface2 .
	InsertFlattenSurface2	Inserts a surface-flatten feature in the model.
	InsertFlexFeature	Inserts a Flex feature using the selected solid or surface body.
	InsertFormToolFeature	Obsolete. Superseded by IFeatureManager.CreateDefinition and IFeatureManager.CreateFeature .
	InsertFreeform	Obsolete. Superseded by IFeatureManager.InsertFreeform2 .
	InsertFreeform2	Inserts a Freeform feature.
	InsertGlobalBoundingBox	Obsolete. See IFeatureManager.CreateDefinition and IBoundingBoxFeatureData .
	InsertGlobalBoundingBox2	Inserts the specified global bounding box using a custom coordinate system. See IFeatureManager.CreateDefinition and IBoundingBoxFeatureData .
	InsertGridFeature	Inserts a Grid System feature.
	InsertGroundPlane	Obsolete. See IFeatureManager.CreateDefinition and IGroundPlaneFeatureData .
	InsertGussetFeature	Obsolete. Superseded by IFeatureManager.InsertGussetFeature3 .
	InsertGussetFeature2	Inserts a gusset feature for the specified faces.
	InsertGussetFeature3	Inserts a gusset feature for pre-selected faces of a weldment.
	InsertIndent	Inserts an indent feature using a selected target body and tool body regions.
	InsertLiveSectionPlane	Inserts a Live Section Plane using the selected plane or planar face.
	InsertMacroFeature	Obsolete. Superseded by IFeatureManager.InsertMacroFeature3 .
	InsertMacroFeature2	Obsolete. Superseded by IFeatureManager.InsertMacroFeature3 .

		
	InsertMacroFeature3	Inserts a macro feature in this model.
	InsertMateReference	Obsolete. Superseded by IFeatureManager.InsertMateReference2 .
	InsertMateReference2	Inserts a mate reference for a part or assembly document.
	InsertMidSurface	Inserts a midsurface feature.
	InsertMirrorFeature	Obsolete. Superseded by IFeatureManager.InsertMirrorFeature2 .
	InsertMirrorFeature2	Mirrors selected features, faces, bodies, and structure systems about a selected plane or planar face.
	InsertMoldCoreCavitySolids	Creates a core/cavity solid feature.
	InsertMoldPartingLine	Inserts a mold parting line feature.
	InsertMoldPartingSurface	Inserts a mold parting surface feature.
	InsertMoldShutOffSurface	Inserts the mold shut-off surface feature.
	InsertMoveCopyBody	Obsolete. Superseded by IFeatureManager.InsertMoveCopyBody2 .
	InsertMoveCopyBody2	Moves or makes copies of the selected solid bodies or surfaces.
	InsertMoveFace	Obsolete. Superseded by IFeatureManager.InsertMoveFace2 and IFeatureManager.InsertMoveFace3 .
	InsertMoveFace2	Obsolete. Superseded by IFeatureManager.InsertMoveFace3 .
	InsertMoveFace3	Moves the selected faces on a solid or surface model.
	InsertMultiFaceDraft	Inserts a multiface draft feature.
	InsertNetBlend	Obsolete. Superseded by IFeatureManager.InsertNetBlend2 .
	InsertNetBlend2	This method inserts a boundary feature or a boundary surface feature.
	InsertProtrusionBlend	Obsolete. Superseded by IFeatureManager.InsertProtrusionBlend2 .
	InsertProtrusionBlend2	Creates a lofted body or boss from the selected profiles, centerline, and guide curves.
	InsertProtrusionSwept	Obsolete. Superseded by IFeatureManager.InsertProtrusionSwept3 .
	InsertProtrusionSwept2	Obsolete. Superseded by IFeatureManager.InsertProtrusionSwept3 .
	InsertProtrusionSwept3	Obsolete. Superseded by IFeatureManager.InsertProtrusionSwept4 .
	InsertProtrusionSwept4	Obsolete. See Remarks .
	InsertReferencePoint	Creates the geometry for the reference points based on any of these selected entities: edges, faces, planes, vertices, or sketch geometry.
	InsertRefPlane	Inserts a constraint-based reference plane using the selected reference entities.
	InsertRevolvedRefSurface	Creates a revolved reference surface by revolving a profile around a centerline.
	InsertRib	Inserts a rib.
	InsertRuledSurfaceFromEdge	Obsolete. Superseded by IFeatureManager.InsertRuledSurfaceFromEdge2 .
	InsertRuledSurfaceFromEdge2	Inserts a ruled surface from the selected edge on this feature.
	InsertSaveOutBodies	Saves the selected surface bodies or solid bodies or sub-weldments to a file.
	InsertScale	Applies the specified scaling to either the current model or a selected graphic body.
	InsertSecurityNote	Inserts a note for the specified macro feature .
	InsertSewRefSurface	Creates a surface by knitting the selected surfaces together.
	InsertSheetMetal3dBend	Inserts a 3D bend in sheet metal part.
	InsertSheetMetalBaseFlange	Obsolete. Superseded by IFeatureManager.InsertSheetMetalBaseFlange2 .
	InsertSheetMetalBaseFlange2	Obsolete. Superseded by IFeatureManager.CreateDefinition and IFeatureManager.CreateFeature .
	InsertSheetMetalCornerTrim	Inserts a break corner trim in the sheet metal part.
	InsertSheetMetalEdgeFlange	Obsolete. Superseded by IFeatureManager.InsertSheetMetalEdgeFlange2 .
	InsertSheetMetalEdgeFlange2	Obsolete. Superseded by IFeatureManager.CreateDefinition and IFeatureManager.CreateFeature .
	InsertSheetMetalGussetFeature	Obsolete. Superseded by IFeatureManager.InsertSheetMetalGussetFeature3 .

	InsertSheetMetalGussetFeature2	Obsolete. Superseded by IFeatureManager::InsertSheetMetalGussetFeature3 .
	InsertSheetMetalGussetFeature3	Obsolete. Superseded by IFeatureManager::CreateDefinition and IFeatureManager::CreateFeature .
	InsertSheetMetalHem	Obsolete. Superseded by IFeatureManager::InsertSheetMetalHem2 .
	InsertSheetMetalHem2	Inserts a hem of the specified relief type at the selected edges of the current sheet metal part.
	InsertSheetMetalLoftedBend	Obsolete. Superseded by IFeatureManager::InsertSheetMetalLoftedBend2 .
	InsertSheetMetalLoftedBend2	Inserts a lofted bend in a sheet metal part.
	InsertSheetMetalMiterFlange	Inserts a meter flange in a sheet metal part.
	InsertSlicing	Creates and inserts slicing into the FeatureManager design tree.
	InsertSplitLineIntersect	Creates split lines using the selected intersecting tool and target entities.
	InsertStructuralWeldment	Obsolete. Superseded by IFeatureManager::InsertStructuralWeldment2 .
	InsertStructuralWeldment2	Obsolete. Superseded by IFeatureManager::InsertStructuralWeldment3 .
	InsertStructuralWeldment3	Obsolete. Superseded by IFeatureManager::InsertStructuralWeldment4 .
	InsertStructuralWeldment4	Obsolete. Superseded by IFeatureManager::InsertStructuralWeldment5 .
	InsertStructuralWeldment5	Inserts a structural weldment feature.
	InsertSubFolder	Creates a subfolder in the Solid Bodies folder in the FeatureManager design tree and moves the selected solid bodies or subfolders in the Solid Bodies folder to the new subfolder.
	InsertSubWeldFolder	Creates a sub weld folder feature containing solid bodies that are pre-selected in the user interface.
	InsertSubWeldFolder2	Creates a sub weld folder feature containing the specified weldment bodies.
	InsertSweepSurface	Obsolete. Superseded by IFeatureManager::InsertSweepSurface2 .
	InsertSweepSurface2	Obsolete. Superseded by IFeatureManager::InsertSweepSurface3 .
	InsertSweepSurface3	Obsolete. See Remarks .
	InsertTableDrivenPattern	Obsolete. Superseded by IFeatureManager::InsertTableDrivenPattern2 .
	InsertTableDrivenPattern2	Obsolete. See IFeatureManager::CreateFeature and the Remarks in ITablePatternFeatureData .
	InsertUntrimSurface	Obsolete. Superseded by IFeatureManager::InsertUntrimSurface2 .
	InsertUntrimSurface2	Extends a surface along its natural boundaries or fills interior surface holes, optionally trimming outside these boundaries or holes.
	InsertVariablePitchHelix	Starts a variable-pitch helix using the selected sketch containing an arc.
	InsertVaryInstanceIncrement	Obsolete. Superseded by IInstanceToVaryOptions .
	InsertVaryInstanceOverride	Obsolete. Superseded by IInstanceToVaryOptions .
	InsertWeldmentCutList	Inserts a cut-list-item folder feature containing pre-selected weldment bodies.
	InsertWeldmentCutList2	Inserts a cut-list-item folder feature containing the specified weldment bodies.
	InsertWeldmentFeature	Inserts a weldment feature.
	InsertWeldmentTrimFeature	Inserts a weldment trim feature.
	InsertWeldmentTrimFeature2	Inserts a weldment trim feature for the specified weldment bodies or faces.
	InsertWrapFeature	Obsolete. Superseded by IFeatureManager::InsertWrapFeature2 .
	InsertWrapFeature2	Inserts a wrap feature.
	IsNameUsed	Checks to see whether the specified name is unique in the FeatureManager design tree and valid to use.
	MakeStyledCurves	Obsolete. Superseded by IFeatureManager::MakeStyledCurves2 .
	MakeStyledCurves2	Fits a spline to the preselected sketch segments to make a smooth edge on the model.
	MoldUndercutDetect	Obsolete. Superseded by IFeatureManager::MoldUndercutDetect2 .
	MoldUndercutDetect2	Detects trapped, also called undercut, areas in a model that cannot be ejected from the mold.
	MoveRotateLiveSectionPlane	Moves or rotates the selected Live Section Plane using the selected Live Section Plane and its manipulator.
	MoveToFolder	Moves the selected feature or folder in the Solid Bodies Feature Manager design tree structure to the specified folder in the Solid Bodies Feature Manager design tree structure.
	PostIntersect	Creates an intersect feature.
	PostSplitBody	Obsolete. Superseded by IFeatureManager::PostSplitBody2 .

		
	PostSplitBody2	Creates a Split feature.
	PostTrimSurface	Sets whether to sew the resulting trimmed surfaces.
	PreIntersect	Obsolete. Superseded by IFeatureManager.PreIntersect2
	PreIntersect2	Prepares an intersect feature.
	PreSplitBody	Obsolete. Superseded by IFeatureManager.PreSplitBody2
	PreSplitBody2	Gets all of the bodies created by splitting a part.
	PreTrimSurface	Sets the trimming options before trimming a surface.
	SetComponentIdentifiers	Allows you to specify the primary, (secondary), and < tertiary > elements to display for the components in the FeatureManager design tree.
	SetFreeformBoundaryContinuity	Sets the boundary continuity for this Freeform feature.
	SetFreeformCurveData	Adds a curve to the pre-selected face for a Freeform feature.
	SetFreeformPointData	Adds a point to a curve for a Freeform feature.
	SetNetBlendCurveData	Sets the data for a curve for this boundary feature or boundary surface feature.
	SetNetBlendDirectionData	Sets the curve set data (one for each of the two directions) for this boundary feature or boundary surface feature.
	ShowBodies	Shows both hidden solid and surface bodies.
	SimpleHole	Obsolete. Superseded by IFeatureManager.SimpleHole2
	SimpleHole2	Inserts a simple hole feature.
	UpdateFeatureTree	Updates the FeatureManager design tree.

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See Also

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IFeatureManager Interface Members

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[SolidWorks.Interop.sldworks Namespace](#) : IFeatureManager Interface

The following tables list the members exposed by [IFeatureManager](#).

Public Properties

Name	Description
 ComponentPrimaryIdentifier	Gets the primary element displayed for the components in the FeatureManager design tree.
 ComponentSecondaryIdentifier	Gets the secondary element(s) displayed for the components in the FeatureManager design tree.
 ComponentTertiaryIdentifier	Gets the tertiary element(s) displayed for the components in the FeatureManager design tree.
 Document	Gets the specified document.
 EnableFeatureTree	Gets or sets whether or not to update the FeatureManager design tree.
 EnableFeatureTreeWindow	Gets or sets whether the FeatureManager design tree is enabled or not.
 FeatureName	Gets the feature name for the specified feature ID.
 FeatureStatistics	Gets statistics about the features in a part document.
 GroupComponentInstances	Gets or sets whether to group the same components in the same configuration in an assembly into a folder in the FeatureManager design tree.
 HideComponentSingleConfigurationOrDisplayStateNames	Gets or sets whether to hide a component's only configuration or display state.
 MoveSizeFeatures	Shows or hides the feature Instant3D.
 ShowComponentConfigurationDescriptions	Gets or sets whether to show the active configuration's component configuration descriptions in the FeatureManager design tree.
 ShowComponentConfigurationNames	Gets or sets whether to show the active configuration's component configuration names in the FeatureManager design tree.

	ShowComponentDescriptions	Gets or sets whether to show the component configuration descriptions in the FeatureManager design tree.
	ShowComponentNames	Gets or sets whether to show the component configuration names in the FeatureManager design tree.
	ShowDisplayStateNames	Gets or sets whether to show the display state names in the FeatureManager design tree.
	ShowFeatureDescription	Gets or sets whether to show the description of the feature in the FeatureManager design tree.
	ShowFeatureDetails	Gets or sets whether to show the feature details in the FeatureManager design tree.
	ShowFeatureName	Gets or sets whether to show the name of the feature in the FeatureManager design tree.
	ShowHierarchyOnly	Gets or sets whether to show only the hierarchy in the FeatureManager design tree.
	SolidForTrim	Gets or sets whether a surface trim feature is a solid body or a surface body.
	ViewDependencies	Gets or sets whether to view the FeatureManager design tree by its dependencies.
	ViewFeatures	Gets or sets whether to view the FeatureManager design tree by its features.

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Public Methods

Name	Description
 AddCornerReliefCorner	Adds the bend corner of two selected faces of a sheet metal body to the set of corners to which to apply a corner relief.
 AddCornerReliefType	Specifies the type of corner relief to apply to the specified corner of the selected sheet metal body.
 AddSMNormalCut	Obsolete. See IFeatureManager.CreateFeature and the Remarks in ISMNormalCutGroupData , and ISMNormalCutFeatureData2 .
 AddSMNormalCutType	Obsolete. See IFeatureManager.CreateFeature and the Remarks in ISMNormalCutGroupData , and ISMNormalCutFeatureData2 .
 AddVariablePitchHelixFirstPitchAndDiameter	Adds the first segment to a variable-pitch helix.
 AddVariablePitchHelixSegment	Adds a segment to a variable-pitch helix.
 AdvancedHole	Obsolete. Superseded by IFeatureManager.AdvancedHole2 .
 AdvancedHole2	Creates an Advanced Hole feature.
 CreateCoordinateSystem	Creates a coordinate system feature using the specified entities.
 CreateCoordinateSystemUsingNumericalValues	Creates a coordinate system feature using the specified numerical values for position and orientation with respect to the global coordinate system.
 CreateCustomBendAllowance	Creates a custom bend allowance to use when creating a sheet metal feature.
 CreateDefinition	Creates a feature data object of the specified type.
 CreateFeature	Creates the specified feature.
 CreateFormTool	Obsolete. Superseded by IFeatureManager.CreateFormTool2 .
 CreateFormTool2	Creates a forming tool feature with the specified point of insertion in a sheet metal part.
 CreateSaveBodyFeature	Creates a Save Bodies feature and creates part and assembly documents of the save bodies.
 CreateStructuralMemberGroup	Creates a weldment structural-member group.
 CutListFolderLocation	Gets the parent folder of the specified solid body / cut list item / body folder.
 DraftXpertChange	Changes the parameters on the selected drafted faces, regardless of whether the drafted faces were created manually or with DraftXpert, provided that DraftXpert can process them.
 DraftXpertRemove	Deletes the draft on the selected faces. If all the faces of a draft are selected, then this method deletes the draft feature; if not, then this method edits the draft feature and removes the selected face references from it.
 EditDeleteFace	Edits a DeleteFace feature.
 EditFreeze	Obsolete. Superseded by IFeatureManager.EditFreeze2 .
 EditFreeze2	Moves the freeze bar to the specified location in the FeatureManager design tree.
 EditReferencePoint	Edits the selected reference points.
 EditRollback	Rolls back or forward the rollback bar to a specific location in the FeatureManager design tree.
 EndVariablePitchHelix	Ends and inserts a variable-pitch helix.
 ExpandFeature	Expands the specified component in the specified FeatureManager design tree pane.
 FeatureAdvancedTableDrivenPattern	Obsolete. See IFeatureManager.CreateFeature and the Remarks in IDimPatternFeatureData .
 FeatureBossThicken	Thickens the selected reference surface, and then generates a boss.

	FeatureChainPattern	Obsolete. See IFeatureManager::CreateFeature and the Remarks in IChainPatternFeatureData .
	FeatureCircularPattern	Obsolete. Superseded by IFeatureManager:FeatureCircularPattern2 .
	FeatureCircularPattern2	Obsolete. Superseded by IFeatureManager:FeatureCircularPattern3 .
	FeatureCircularPattern3	Obsolete. Superseded by IFeatureManager:FeatureCircularPattern4 .
	FeatureCircularPattern4	Obsolete. Superseded by IFeatureManager:FeatureCircularPattern5 .
	FeatureCircularPattern5	Obsolete. See IFeatureManager::CreateFeature and the Remarks in ICircularPatternFeatureData and ILocalCircularPatternFeatureData .
	FeatureCut	Obsolete. Superseded by IFeatureManager:FeatureCut2 .
	FeatureCut2	Obsolete. Superseded by IFeatureManager:FeatureCut3 .
	FeatureCut3	Obsolete. Superseded by IFeatureManager:FeatureCut4 .
	FeatureCut4	Creates a cut extrude feature.
	FeatureCutThicken	Thickens the selected reference surface feature, and then generates a cut.
	FeatureCutThin	Obsolete. Superseded by IFeatureManager:FeatureCutThin2 .
	FeatureCutThin2	Inserts a thin cut extrude feature.
	FeatureDimensionPattern	Not implemented.
	FeatureExtruRefSurface	Obsolete. Superseded by IFeatureManager:FeatureExtruRefSurface2 .
	FeatureExtruRefSurface2	Obsolete. Superseded by IFeatureManager:FeatureExtruRefSurface3 .
	FeatureExtruRefSurface3	Inserts an extruded surface.
	FeatureExtrusion	Obsolete. Superseded by IFeatureManager:FeatureExtrusion2 .
	FeatureExtrusion2	Obsolete. Superseded by IFeatureManager:FeatureExtrusion3 .
	FeatureExtrusion3	Creates an extruded feature.
	FeatureExtrusionThin	Obsolete. Superseded by IFeatureManager:FeatureExtrusionThin2 .
	FeatureExtrusionThin2	Creates an extruded thin feature.
	FeatureFillet	Obsolete. Superseded by IFeatureManager:FeatureFillet2 .
	FeatureFillet2	Obsolete. Superseded by IFeatureManager:FeatureFillet3 .
	FeatureFillet3	Creates a fillet feature for selected edges and control point references.
	FeatureFillPattern	Obsolete. See IFeatureManager::CreateFeature and the Remarks in IFillPatternFeatureData .
	FeatureFolderLocation	Gets the folder feature for the specified feature.
	FeatureLinearPattern	Obsolete. Superseded by IFeatureManager:FeatureLinearPattern2 .
	FeatureLinearPattern2	Obsolete. Superseded by IFeatureManager:FeatureLinearPattern3 .
	FeatureLinearPattern3	Obsolete. Superseded by IFeatureManager:FeatureLinearPattern4 .
	FeatureLinearPattern4	Obsolete. Superseded by IFeatureManager:FeatureLinearPattern5 .
	FeatureLinearPattern5	Obsolete. See IFeatureManager::CreateFeature and the Remarks in ILinearPatternFeatureData and ILocalLinearPatternFeatureData .
	FeatureLocalCurveDrivenPattern	Obsolete. See IFeatureManager::CreateFeature and the Remarks in ILocalCurvePatternFeatureData .
	FeatureLocalSketchDrivenPattern	Obsolete. See IFeatureManager::CreateFeature and the Remarks in ILocalSketchPatternFeatureData .
	FeatureRevolve	Obsolete. Superseded by IFeatureManager:FeatureRevolve2 .
	FeatureRevolve2	Creates a base-, boss-, or cut-revolve feature.
	FeatureRevolveCut	Obsolete. Superseded by IFeatureManager:FeatureRevolveCut2 .
	FeatureRevolveCut2	Obsolete. Superseded by IFeatureManager:FeatureRevolve2 .
	FeatureRevolveThin	Obsolete. Superseded by IFeatureManager:FeatureRevolve2 .
	FeatureRevolveThinCut	Obsolete. Superseded by IFeatureManager:FeatureRevolve2 .
	FeatureSketchDrivenPattern	Obsolete. See IFeatureManager::CreateFeature and the Remarks in ISketchPatternFeatureData .

	FilletXpertChange	Changes the parameters on the selected filleted faces, regardless of whether the filleted faces were created manually or with FilletXpert, provided that FilletXpert can process them.
	FilletXpertMakeCorner	Creates or changes a fillet corner feature.
	FilletXpertRemove	Deletes the fillets on the selected faces.
	FinishCornerRelief	Creates a sheet metal corner relief feature.
	FinishSMNormalCut	Obsolete. See IFeatureManager::CreateFeature and the Remarks in ISMNormalCutGroupData , and ISMNormalCutFeatureData2 .
	GetCornerManagementFolders	Gets the structure system corner management folders.
	GetCreateFeatureErrors	Gets the errors that occurred during the last call to IFeatureManager::CreateFeature .
	GetFeatureCount	Gets the number of features in this document.
	GetFeatures	Gets the features in this document.
	GetFeatureTreeRootItem	Obsolete. Superseded by IFeatureManager::GetFeatureTreeRootItem2 .
	GetFeatureTreeRootItem2	Gets the root item of the FeatureManager design tree in the specified pane.
	GetFlatPatternFolder	Gets the interface to the flat-pattern folder feature in the FeatureManager design tree.
	GetFreezeLocation	Gets the location of the freeze bar in the FeatureManager design tree.
	GetPreTrimmedBodies	Gets the temporary trimmed bodies using the specified target sheet (surface) body according to the trim tools previously defined by IFeatureManager::PreTrimSurface .
	GetSelectionSetFolder	Gets the Selection Sets folder.
	GetSheetMetalFolder	Gets the interface to the sheet metal folder feature in the FeatureManager design tree.
	GetSlicingData	Gets a new slicing data object with default parameter values.
	GetStructureSystemFolders	Gets the structure system folders.
	HideBodies	Hides both solid and surface bodies in the model.
	HoleWizard	Obsolete. Superseded by IFeatureManager::HoleWizard2 .
	HoleWizard2	Obsolete. Superseded by IFeatureManager::HoleWizard3 .
	HoleWizard3	Obsolete. Superseded by IFeatureManager::HoleWizard4 .
	HoleWizard4	Obsolete. Superseded by IFeatureManager::HoleWizard5 .
	HoleWizard5	Creates customized holes of various kinds.
	IFeatureFillet	Obsolete. Superseded by IFeatureManager::IFeatureFillet2 .
	IFeatureFillet2	Obsolete. Superseded by IFeatureManager::FeatureFillet3 .
	IGetFeatures	Gets the features in this document.
	InsertCombineFeature	Combines the specified bodies in the multibody part to create a combine feature.
	InsertMacroFeature	Obsolete. Superseded by IFeatureManager::InsertMacroFeature3 .
	InsertMacroFeature2	Obsolete. Superseded by IFeatureManager::InsertMacroFeature3 .
	InsertMacroFeature3	Inserts a macro feature in this model.
	InsertMoveFace2	Obsolete. Superseded by IFeatureManager::InsertMoveFace3 .
	InsertReferencePoint	Creates the geometry for the reference points based on any of these selected entities: edges, faces, planes, vertices, or sketch geometry.
	InsertSheetMetalEdgeFlange2	Obsolete. Superseded by IFeatureManager::CreateDefinition and IFeatureManager::CreateFeature .
	InsertTableDrivenPattern	Obsolete. Superseded by IFeatureManager::InsertTableDrivenPattern2 .
	InsertCenterOfMass	Inserts a Center of Mass feature.
	InsertCenterOfMassReferencePoint	Inserts a Center of Mass Reference Point feature.
	InsertCombineFeature	Combines the specified bodies in the multibody part to create a combine feature.
	InsertConnectionPoint	Adds a connection point based on the selected entities.
	InsertConvertToSheetMetal	Obsolete. Superseded by IFeatureManager::InsertConvertToSheetMetal2 .
	InsertConvertToSheetMetal2	Obsolete. Superseded by IFeatureManager::CreateDefinition and IFeatureManager::CreateFeature .
	InsertCoordinateSystem	Inserts a coordinate system feature.
	InsertCosmeticThread	Obsolete. Superseded by IFeatureManager::InsertCosmeticThread2 .

		
	InsertCosmeticThread2	Obsolete. Superseded by IFeatureManager.InsertCosmeticThread3 .
	InsertCosmeticThread3	Inserts a cosmetic thread.
	InsertCosmeticWeldBead	Obsolete. Superseded by IFeatureManager.InsertCosmeticWeldBead2 .
	InsertCosmeticWeldBead2	Inserts a cosmetic weld bead using either weld geometry or a weld path.
	InsertCrossBreak	Inserts a cross break feature on the selected face in a sheet metal part.
	InsertCutBlend	Inserts a lofted cut based on the selected profiles, centerline, and guide curves.
	InsertCutSurface	Inserts a surface-cut feature using the preselected surface or plane.
	InsertCutSwept	Obsolete. Superseded by IFeatureManager.InsertCutSwept3 .
	InsertCutSwept2	Obsolete. Superseded by IFeatureManager.InsertCutSwept3 .
	InsertCutSwept3	Obsolete. Superseded by IFeatureManager.InsertCutSwept4 .
	InsertCutSwept4	Obsolete. Superseded by IFeatureManager.InsertCutSwept5 .
	InsertCutSwept5	Obsolete. See Remarks .
	InsertDeleteBody	Obsolete. Superseded by IFeatureManager.InsertDeleteBody2 .
	InsertDeleteBody2	Inserts a Body-Delete/Keep feature.
	InsertDeleteHoleForSurface	Inserts a Delete Hole feature for one or more selected hole edges on a surface.
	InsertDerivedPattern	Obsolete. Superseded by IFeatureManager.InsertDerivedPattern2 .
	InsertDerivedPattern2	Obsolete. See IFeatureManager.CreateFeature and the Remarks in IDerivedPatternFeatureData .
	InsertDwgOrDxfFile	Obsolete. Superseded by IFeatureManager.InsertDwgOrDxfFile2 .
	InsertDwgOrDxfFile2	Inserts a DXF/DWG image feature.
	InsertEdgeMerge	Merges multiple edges into a single edge using the selected faces when importing data.
	InsertEndCapFeature	Obsolete. Superseded by IFeatureManager.InsertEndCapFeature3 .
	InsertEndCapFeature2	Inserts an end cap feature using the specified end faces of a structural member.
	InsertEndCapFeature3	Inserts an end cap feature for one or more pre-selected open ends of a structural member.
	InsertFeatureChamfer	Inserts a chamfer.
	InsertFeatureLock	Locks or freezes a selected feature.
	InsertFeatureTreeFolder	Obsolete. Superseded by IFeatureManager.InsertFeatureTreeFolder2 .
	InsertFeatureTreeFolder2	Inserts a folder in the FeatureManager design tree for pre-selected features or components.
	InsertFilletBeadFeature	Obsolete. Superseded by IFeatureManager.InsertFilletBeadFeature2 .
	InsertFilletBeadFeature2	Inserts a fillet weld bead feature and also fills the gap between the pre-selected part bodies, if any.
	InsertFilletBeadFeature3	Inserts fillet weld bead features for the specified face sets.
	InsertFillSurface	Obsolete. Superseded by IFeatureManager.InsertFillSurface2 .
	InsertFillSurface2	Inserts a fill-surface feature in the model.
	InsertFlattenSurface	Obsolete. Superseded by IFeatureManager.InsertFlattenSurface2 .
	InsertFlattenSurface2	Inserts a surface-flatten feature in the model.
	InsertFlexFeature	Inserts a Flex feature using the selected solid or surface body.
	InsertFormToolFeature	Obsolete. Superseded by IFeatureManager.CreateDefinition and IFeatureManager.CreateFeature .
	InsertFreeform	Obsolete. Superseded by IFeatureManager.InsertFreeform2 .
	InsertFreeform2	Inserts a Freeform feature.
	InsertGlobalBoundingBox	Obsolete. See IFeatureManager.CreateDefinition and IBoundingBoxFeatureData .
	InsertGlobalBoundingBox2	Inserts the specified global bounding box using a custom coordinate system. See IFeatureManager.CreateDefinition and IBoundingBoxFeatureData .
	InsertGridFeature	Inserts a Grid System feature.
	InsertGroundPlane	Obsolete. See IFeatureManager.CreateDefinition and IGroundPlaneFeatureData .

		
	InsertGussetFeature	Obsolete. Superseded by IFeatureManager.InsertGussetFeature3 .
	InsertGussetFeature2	Inserts a gusset feature for the specified faces.
	InsertGussetFeature3	Inserts a gusset feature for pre-selected faces of a weldment.
	InsertIndent	Inserts an indent feature using a selected target body and tool body regions.
	InsertLiveSectionPlane	Inserts a Live Section Plane using the selected plane or planar face.
	InsertMacroFeature	Obsolete. Superseded by IFeatureManager.InsertMacroFeature3 .
	InsertMacroFeature2	Obsolete. Superseded by IFeatureManager.InsertMacroFeature3 .
	InsertMacroFeature3	Inserts a macro feature in this model.
	InsertMateReference	Obsolete. Superseded by IFeatureManager.InsertMateReference2 .
	InsertMateReference2	Inserts a mate reference for a part or assembly document.
	InsertMidSurface	Inserts a midsurface feature.
	InsertMirrorFeature	Obsolete. Superseded by IFeatureManager.InsertMirrorFeature2 .
	InsertMirrorFeature2	Mirrors selected features, faces, bodies, and structure systems about a selected plane or planar face.
	InsertMoldCoreCavitySolids	Creates a core/cavity solid feature.
	InsertMoldPartingLine	Inserts a mold parting line feature.
	InsertMoldPartingSurface	Inserts a mold parting surface feature.
	InsertMoldShutOffSurface	Inserts the mold shut-off surface feature.
	InsertMoveCopyBody	Obsolete. Superseded by IFeatureManager.InsertMoveCopyBody2 .
	InsertMoveCopyBody2	Moves or makes copies of the selected solid bodies or surfaces.
	InsertMoveFace	Obsolete. Superseded by IFeatureManager.InsertMoveFace2 and IFeatureManager.InsertMoveFace3 .
	InsertMoveFace2	Obsolete. Superseded by IFeatureManager.InsertMoveFace3 .
	InsertMoveFace3	Moves the selected faces on a solid or surface model.
	InsertMultiFaceDraft	Inserts a multiface draft feature.
	InsertNetBlend	Obsolete. Superseded by IFeatureManager.InsertNetBlend2 .
	InsertNetBlend2	This method inserts a boundary feature or a boundary surface feature.
	InsertProtrusionBlend	Obsolete. Superseded by IFeatureManager.InsertProtrusionBlend2 .
	InsertProtrusionBlend2	Creates a lofted body or boss from the selected profiles, centerline, and guide curves.
	InsertProtrusionSwept	Obsolete. Superseded by IFeatureManager.InsertProtrusionSwept3 .
	InsertProtrusionSwept2	Obsolete. Superseded by IFeatureManager.InsertProtrusionSwept3 .
	InsertProtrusionSwept3	Obsolete. Superseded by IFeatureManager.InsertProtrusionSwept4 .
	InsertProtrusionSwept4	Obsolete. See Remarks .
	InsertReferencePoint	Creates the geometry for the reference points based on any of these selected entities: edges, faces, planes, vertices, or sketch geometry.
	InsertRefPlane	Inserts a constraint-based reference plane using the selected reference entities.
	InsertRevolvedRefSurface	Creates a revolved reference surface by revolving a profile around a centerline.
	InsertRib	Inserts a rib.
	InsertRuledSurfaceFromEdge	Obsolete. Superseded by IFeatureManager.InsertRuledSurfaceFromEdge2 .
	InsertRuledSurfaceFromEdge2	Inserts a ruled surface from the selected edge on this feature.
	InsertSaveOutBodies	Saves the selected surface bodies or solid bodies or sub-weldments to a file.
	InsertScale	Applies the specified scaling to either the current model or a selected graphic body.
	InsertSecurityNote	Inserts a note for the specified macro feature .
	InsertSewRefSurface	Creates a surface by knitting the selected surfaces together.

	InsertSheetMetal3dBend	Inserts a 3D bend in sheet metal part.
	InsertSheetMetalBaseFlange	Obsolete. Superseded by IFeatureManager::InsertSheetMetalBaseFlange2 .
	InsertSheetMetalBaseFlange2	Obsolete. Superseded by IFeatureManager::CreateDefinition and IFeatureManager::CreateFeature .
	InsertSheetMetalCornerTrim	Inserts a break corner trim in the sheet metal part.
	InsertSheetMetalEdgeFlange	Obsolete. Superseded by IFeatureManager::InsertSheetMetalEdgeFlange2 .
	InsertSheetMetalEdgeFlange2	Obsolete. Superseded by IFeatureManager::CreateDefinition and IFeatureManager::CreateFeature .
	InsertSheetMetalGussetFeature	Obsolete. Superseded by IFeatureManager::InsertSheetMetalGussetFeature3 .
	InsertSheetMetalGussetFeature2	Obsolete. Superseded by IFeatureManager::InsertSheetMetalGussetFeature3 .
	InsertSheetMetalGussetFeature3	Obsolete. Superseded by IFeatureManager::CreateDefinition and IFeatureManager::CreateFeature .
	InsertSheetMetalHem	Obsolete. Superseded by IFeatureManager::InsertSheetMetalHem2 .
	InsertSheetMetalHem2	Inserts a hem of the specified relief type at the selected edges of the current sheet metal part.
	InsertSheetMetalLoftedBend	Obsolete. Superseded by IFeatureManager::InsertSheetMetalLoftedBend2 .
	InsertSheetMetalLoftedBend2	Inserts a lofted bend in a sheet metal part.
	InsertSheetMetalMiterFlange	Inserts a meter flange in a sheet metal part.
	InsertSlicing	Creates and inserts slicing into the FeatureManager design tree.
	InsertSplitLineIntersect	Creates split lines using the selected intersecting tool and target entities.
	InsertStructuralWeldment	Obsolete. Superseded by IFeatureManager::InsertStructuralWeldment2 .
	InsertStructuralWeldment2	Obsolete. Superseded by IFeatureManager::InsertStructuralWeldment3 .
	InsertStructuralWeldment3	Obsolete. Superseded by IFeatureManager::InsertStructuralWeldment4 .
	InsertStructuralWeldment4	Obsolete. Superseded by IFeatureManager::InsertStructuralWeldment5 .
	InsertStructuralWeldment5	Inserts a structural weldment feature.
	InsertSubFolder	Creates a subfolder in the Solid Bodies folder in the FeatureManager design tree and moves the selected solid bodies or subfolders in the Solid Bodies folder to the new subfolder.
	InsertSubWeldFolder	Creates a sub weld folder feature containing solid bodies that are pre-selected in the user interface.
	InsertSubWeldFolder2	Creates a sub weld folder feature containing the specified weldment bodies.
	InsertSweepSurface	Obsolete. Superseded by IFeatureManager::InsertSweepSurface2 .
	InsertSweepSurface2	Obsolete. Superseded by IFeatureManager::InsertSweepSurface3 .
	InsertSweepSurface3	Obsolete. See Remarks .
	InsertTableDrivenPattern	Obsolete. Superseded by IFeatureManager::InsertTableDrivenPattern2 .
	InsertTableDrivenPattern2	Obsolete. See IFeatureManager::CreateFeature and the Remarks in ITablePatternFeatureData .
	InsertUntrimSurface	Obsolete. Superseded by IFeatureManager::InsertUntrimSurface2 .
	InsertUntrimSurface2	Extends a surface along its natural boundaries or fills interior surface holes, optionally trimming outside these boundaries or holes.
	InsertVariablePitchHelix	Starts a variable-pitch helix using the selected sketch containing an arc.
	InsertVaryInstanceIncrement	Obsolete. Superseded by IInstanceToVaryOptions .
	InsertVaryInstanceOverride	Obsolete. Superseded by IInstanceToVaryOptions .
	InsertWeldmentCutList	Inserts a cut-list-item folder feature containing pre-selected weldment bodies.
	InsertWeldmentCutList2	Inserts a cut-list-item folder feature containing the specified weldment bodies.
	InsertWeldmentFeature	Inserts a weldment feature.
	InsertWeldmentTrimFeature	Inserts a weldment trim feature.
	InsertWeldmentTrimFeature2	Inserts a weldment trim feature for the specified weldment bodies or faces.
	InsertWrapFeature	Obsolete. Superseded by IFeatureManager::InsertWrapFeature2 .
	InsertWrapFeature2	Inserts a wrap feature.
	IsNameUsed	Checks to see whether the specified name is unique in the FeatureManager design tree and valid to use.
	MakeStyledCurves	Obsolete. Superseded by IFeatureManager::MakeStyledCurves2 .

		
	MakeStyledCurves2	Fits a spline to the preselected sketch segments to make a smooth edge on the model.
	MoldUndercutDetect	Obsolete. Superseded by IFeatureManager.MoldUndercutDetect2 .
	MoldUndercutDetect2	Detects trapped, also called undercut, areas in a model that cannot be ejected from the mold.
	MoveRotateLiveSectionPlane	Moves or rotates the selected Live Section Plane using the selected Live Section Plane and its manipulator.
	MoveToFolder	Moves the selected feature or folder in the Solid Bodies Feature Manager design tree structure to the specified folder in the Solid Bodies Feature Manager design tree structure.
	PostIntersect	Creates an intersect feature.
	PostSplitBody	Obsolete. Superseded by IFeatureManager.PostSplitBody2 .
	PostSplitBody2	Creates a Split feature.
	PostTrimSurface	Sets whether to sew the resulting trimmed surfaces.
	PreIntersect	Obsolete. Superseded by IFeatureManager.PreIntersect2 .
	PreIntersect2	Prepares an intersect feature.
	PreSplitBody	Obsolete. Superseded by IFeatureManager.PreSplitBody2 .
	PreSplitBody2	Gets all of the bodies created by splitting a part.
	PreTrimSurface	Sets the trimming options before trimming a surface.
	SetComponentIdentifiers	Allows you to specify the primary, (secondary), and < tertiary > elements to display for the components in the FeatureManager design tree.
	SetFreeformBoundaryContinuity	Sets the boundary continuity for this Freeform feature.
	SetFreeformCurveData	Adds a curve to the pre-selected face for a Freeform feature.
	SetFreeformPointData	Adds a point to a curve for a Freeform feature.
	SetNetBlendCurveData	Sets the data for a curve for this boundary feature or boundary surface feature.
	SetNetBlendDirectionData	Sets the curve set data (one for each of the two directions) for this boundary feature or boundary surface feature.
	ShowBodies	Shows both hidden solid and surface bodies.
	SimpleHole	Obsolete. Superseded by IFeatureManager.SimpleHole2 .
	SimpleHole2	Inserts a simple hole feature.
	UpdateFeatureTree	Updates the FeatureManager design tree.

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See Also

[IFeatureManager Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

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IFeatureManager Interface Properties

[See Also](#)

[SolidWorks.Interop.sldworks Namespace](#) : IFeatureManager Interface

For a list of all members of this type, see [IFeatureManager members](#).

Public Properties

Name	Description
 ComponentPrimaryIdentifier	Gets the primary element displayed for the components in the FeatureManager design tree.
 ComponentSecondaryIdentifier	Gets the secondary element(s) displayed for the components in the FeatureManager design tree.
 ComponentTertiaryIdentifier	Gets the tertiary element(s) displayed for the components in the FeatureManager design tree.
 Document	Gets the specified document.
 EnableFeatureTree	Gets or sets whether or not to update the FeatureManager design tree.
 EnableFeatureTreeWindow	Gets or sets whether the FeatureManager design tree is enabled or not.

	FeatureName	Gets the feature name for the specified feature ID.
	FeatureStatistics	Gets statistics about the features in a part document.
	GroupComponentInstances	Gets or sets whether to group the same components in the same configuration in an assembly into a folder in the FeatureManager design tree.
	HideComponentSingleConfigurationOrDisplayStateNames	Gets or sets whether to hide a component's only configuration or display state.
	MoveSizeFeatures	Shows or hides the feature Instant3D.
	ShowComponentConfigurationDescriptions	Gets or sets whether to show the active configuration's component configuration descriptions in the FeatureManager design tree.
	ShowComponentConfigurationNames	Gets or sets whether to show the active configuration's component configuration names in the FeatureManager design tree.
	ShowComponentDescriptions	Gets or sets whether to show the component configuration descriptions in the FeatureManager design tree.
	ShowComponentNames	Gets or sets whether to show the component configuration names in the FeatureManager design tree.
	ShowDisplayStateNames	Gets or sets whether to show the display state names in the FeatureManager design tree.
	ShowFeatureDescription	Gets or sets whether to show the description of the feature in the FeatureManager design tree.
	ShowFeatureDetails	Gets or sets whether to show the feature details in the FeatureManager design tree.
	ShowFeatureName	Gets or sets whether to show the name of the feature in the FeatureManager design tree.
	ShowHierarchyOnly	Gets or sets whether to show only the hierarchy in the FeatureManager design tree.
	SolidForTrim	Gets or sets whether a surface trim feature is a solid body or a surface body.
	ViewDependencies	Gets or sets whether to view the FeatureManager design tree by its dependencies.
	ViewFeatures	Gets or sets whether to view the FeatureManager design tree by its features.

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 See Also

[IFeatureManager Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

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AddCornerReliefCorner Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : AddCornerReliefCorner Method (IFeatureManager)

Adds the bend corner of two selected faces of a sheet metal body to the set of corners to which to apply a corner relief.

 .NET Syntax

Visual Basic (Declaration)

Function AddCornerReliefCorner() As System.Integer

Visual Basic (Usage)



```
Dim instance As IFeatureManager
Dim value As System.Integer

value = instance.AddCornerReliefCorner()
```

C#

System.int AddCornerReliefCorner()

C++/CLI

System.int AddCornerReliefCorner();

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Index of corner to which to apply the corner relief; used by [IFeatureManager::AddCornerReliefType](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::AddCornerReliefCorner](#).

Example

[Create Corner Relief Feature \(C#\)](#)
[Create Corner Relief Feature \(VBA\)](#)
[Create Corner Relief Feature \(VB.NET\)](#)

Remarks

To create a corner relief feature:

1. Call [IModelDocExtension::SelectByID2](#) with Mark = 0 and Append = true to select the sheet metal body in which to create a corner relief feature.
2. Call [IModelDocExtension::SelectByID2](#) with Mark = 4 and Append = true to select two faces that form a bend corner.
3. Call this method to add the corner to the corner relief feature.
4. Call [IFeatureManager::AddCornerReliefType](#) to specify the corner relief for the corner.
5. Repeat steps 2 - 4 to add another corner to the corner relief feature.
6. Call [IFeatureManager::FinishCornerRelief](#).

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IFlatPatternFeatureData Interface](#)

Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

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AddCornerReliefType Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : AddCornerReliefType Method (IFeatureManager)

Specifies the type of corner relief to apply to the specified corner of the selected sheet metal body.

.NET Syntax

Visual Basic (Declaration)

```
Function AddCornerReliefType( _  
    ByVal CornerIndex As System.Integer, _  
    ByVal ReliefType As System.Integer, _  
    ByVal Length1 As System.Double, _  
    ByVal Length2 As System.Double, _  
    ByVal Length3 As System.Double, _  
    ByVal CenterOnBendLines As System.Boolean, _  
    ByVal RatioToThickness As System.Boolean, _  
    ByVal TangentToBend As System.Boolean, _  
    ByVal AddFilletedCorners As System.Boolean, _  
    ByVal NarrowCorner As System.Boolean _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim CornerIndex As System.Integer  
Dim ReliefType As System.Integer  
Dim Length1 As System.Double  
Dim Length2 As System.Double  
Dim Length3 As System.Double  
Dim CenterOnBendLines As System.Boolean  
Dim RatioToThickness As System.Boolean  
Dim TangentToBend As System.Boolean  
Dim AddFilletedCorners As System.Boolean  
Dim NarrowCorner As System.Boolean  
Dim value As System.Boolean
```

```
value = instance.AddCornerReliefType(CornerIndex, ReliefType, Length1, Length2, Length3,
CenterOnBendLines, RatioToThickness, TangentToBend, AddFillettedCorners, NarrowCorner)
```

C#

```
System.bool AddCornerReliefType(
    System.int CornerIndex,
    System.int ReliefType,
    System.double Length1,
    System.double Length2,
    System.double Length3,
    System.bool CenterOnBendLines,
    System.bool RatioToThickness,
    System.bool TangentToBend,
    System.bool AddFillettedCorners,
    System.bool NarrowCorner
)
```

C++/CLI

```
System.bool AddCornerReliefType(
    & System.int CornerIndex,
    & System.int ReliefType,
    & System.double Length1,
    & System.double Length2,
    & System.double Length3,
    & System.bool CenterOnBendLines,
    & System.bool RatioToThickness,
    & System.bool TangentToBend,
    & System.bool AddFillettedCorners,
    & System.bool NarrowCorner
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

CornerIndex
Index of corner to which to apply the corner relief; -1 to apply it to the corner last added with [IFeatureManager::AddCornerReliefCorner](#)

ReliefType
Type of corner relief as defined by [swCornerReliefType_e](#)

Length1
Not used

Length2

If ReliefType is swCornerReliefType_e...	Then Length2 is the slot...
swCornerSquareRelief	Length of the corner relief
swCornerObroundRelief	Length of the corner relief
swCornerCircularRelief	Radius of the corner relief

Length3

If ReliefType is swCornerReliefType_e...	Then Length3 is...
swCornerObroundRelief	Slot width of the corner relief
swCornerSquareRelief and FillettedCorners = true	Radius of filleted corner

CenterOnBendLines

True to center the corner relief relative to the bend lines, false to not; valid only if ReliefType is one of the following:

- swCornerReliefType_e.swCornerSquareRelief
- swCornerReliefType_e.swCornerCircularRelief
- swCornerReliefType_e.swCornerObroundRelief

RatioToThickness

True to use a ratio value to cut the bend area so that the body can be folded, false to not; the ratios for valid relief types are calculated as follows where t = thickness of sheet metal:

If ReliefType is swCornerReliefType_e...	Then ratios are...
swCornerSquareRelief	Length2/t
swCornerCircularRelief	Length2/t
swCornerObroundRelief	Length2/t and Length3/t

TangentToBend

True to make the corner relief tangent to the inside bend edges, false to not

AddFillettedCorners

True to fillet the corner relief corners, false to not; valid only if ReliefType = swCornerReliefType_e.swCornerSquareRelief

NarrowCorner

True to use the algorithm for large bend radii to narrow the corner relief in the bend area, false to not; valid only if ReliefType is one of the following:

- swCornerReliefType_e.swCornerSquareRelief
- swCornerReliefType_e.swCornerCircularRelief
- swCornerReliefType_e.swCornerOroundRelief

Return Value

True if successful, false if not

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::AddCornerReliefType](#).

Example

[Create Corner Relief Feature \(C#\)](#)
[Create Corner Relief Feature \(VBA\)](#)
[Create Corner Relief Feature \(VB.NET\)](#)

Remarks

To create a corner relief feature:

1. Call [IModelDocExtension::SelectById2](#) with Mark = 0 and Append = true to select the sheet metal body in which to create a corner relief feature.
2. Call [IModelDocExtension::SelectById2](#) with Mark = 4 and Append = true to select two faces that form a bend corner.
3. Call [IFeatureManager::AddCornerReliefCorner](#) to add the corner to the corner relief feature.
4. Call this method to specify the corner relief for the corner.
5. Repeat steps 2 - 4 to add another corner to the corner relief feature.
6. Call [IFeatureManager::FinishCornerRelief](#).

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IFlatPatternFeatureData Interface](#)

Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

AddSMNormalCut Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : AddSMNormalCut Method (IFeatureManager)

Obsolete. See [IFeatureManager::CreateFeature](#) and the Remarks in [ISMNormalCutGroupData](#), and [ISMNormalCutFeatureData2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function AddSMNormalCut() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim value As System.Integer  
  
value = instance.AddSMNormalCut()
```

C#

```
System.int AddSMNormalCut()
```

C++/CLI

```
System.int AddSMNormalCut();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Result code as defined in [swSMNormalCutError_e](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::AddSMNormalCut](#).

Remarks

To create a sheet metal Normal Cut feature, see the [IFeatureManager::AddSMNormalCutType](#) Remarks and Example sections.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IFeatureManager::FinishSMNormalCut Method \(\)](#)
[ISMNormalCutFeatureData Interface](#)

Availability

SOLIDWORKS 2018 FCS, Revision Number 26.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

AddSMNormalCutType Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : AddSMNormalCutType Method (IFeatureManager)

Obsolete. See [IFeatureManager::CreateFeature](#) and the Remarks in [ISMNormalCutGroupData](#), and [ISMNormalCutFeatureData2](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub AddSMNormalCutType( _  
    ByVal AutoPropagation As System.Boolean, _  
    ByVal Optimize As System.Boolean, _  
    ByVal BExtent As System.Boolean _  
)
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim AutoPropagation As System.Boolean  
Dim Optimize As System.Boolean  
Dim BExtent As System.Boolean  
  
instance.AddSMNormalCutType(AutoPropagation, Optimize, BExtent)
```

C#

```
void AddSMNormalCutType(  
    System.bool AutoPropagation,  
    System.bool Optimize,  
    System.bool BExtent  
)
```

C++/CLI

```
void AddSMNormalCutType(  
& System.bool AutoPropagation,  
& System.bool Optimize,  
& System.bool BExtent  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

AutoPropagation

True to automatically propagate the Normal Cut, false to not
Optimize
True to optimize geometry, false to not
BExtent
True to maximize the cutout using the top and bottom intersection profiles, false to use an offset plane

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::AddSMNormalCutType](#).

Example

```
'VBA

'1. Open a sheet metal part that has two cut extrudes.
'2. Modify the macro below as needed.

Dim swApp As SldWorks.SldWorks
Dim Part As SldWorks.ModelDoc2
Dim myFeature As SldWorks.Feature
Dim myNormalCutFeatData As SldWorks.SMNormalCutFeatureData
Dim boolstatus As Boolean

Option Explicit
Sub main()

    Set swApp = Application.SldWorks
    Set Part = swApp.ActiveDoc

    'Select a cut-extrude face for which to create a Normal Cut
    boolstatus = Part.Extension.SelectByRay(-6.38655841527864E-02, -1.56055561574817E-02,
3.58410081097382E-03, -9.80322952634684E-02, -0.994744620924993, -2.95433279467378E-02,
2.7317052985792E-04, 2, False, 1, 0)
    Dim OneFaceGroup As Long
    OneFaceGroup = Part.FeatureManager.AddSMNormalCut()
    Part.ClearSelection2 True

    'Use an offset plane to create the Normal Cut feature
    boolstatus = Part.Extension.SelectByRay(-6.00249578326526E-02, -1.83129588701831E-02,
1.70249820822459E-02, -9.80322952634684E-02, -0.994744620924993, -2.95433279467378E-02,
2.69645748827495E-04, 2, False, 4, 0)

    Part.FeatureManager.AddSMNormalCutType True, True, False

    'Create a Normal Cut feature using the previously selected cut-extrude faces
    Set myFeature = Part.FeatureManager.FinishSMNormalCut()

    Set swFeatData = myFeature.GetDefinition
    Dim Number As Long
    Number = swFeatData.GetNumberOfFaceGroups()

    'Select another cut-extrude face for which to create a second Normal Cut
    boolstatus = swFeatData.AccessSelections(Part, Nothing)

    boolstatus = Part.Extension.SelectByRay(-6.24756020285346E-02, -1.97848124440725E-02,
1.18685503457829E-02, -9.80322952634684E-02, -0.994744620924993, -2.95433279467378E-02,
2.7317052985792E-04, 2, False, 1, 0)
    Dim instance As ISelectionMgr
    Set face = Part.SelectionManager.GetSelectedObject6(1, -1)
    Dim faces As Variant
    Dim aFaces() As SldWorks.face
    ReDim aFaces(0)
    Set aFaces(0) = face
    faces = aFaces
    longwarnings = swFeatData.AddAGroupOfFaces(faces, longwarnings)

    Number = swFeatData.GetNumberOfFaceGroups()

    'Change the profile of the Normal Cut by adjusting the offset of the offset plane
    myNormalCutFeatData.Offset = 1.0
```

```

Dim bStatus As Boolean
bStatus = myFeature.ModifyDefinition(swFeatData, Part, Nothing)
Part.ClearSelection2 True

```

End Sub

Remarks

To create a Normal Cut feature in a sheet metal part, call:

1. [IFeatureManager::AddSMNormalCut](#).
2. [IFeatureManager::AddSMNormalCutType](#).
3. [IModelDocExtension::SelectByID2](#) with Mark = 1 to select the cut-extrude faces that you wish to make Normal Cuts.
4. [IModelDocExtension::SelectByID2](#) with Mark = 2 to select the group of cut-extrude faces.
5. [IModelDocExtension::SelectByID2](#) with Mark = 4 to select the offset plane, if BExtent is false.
6. (Optional) [IModelDocExtension::SelectByID2](#) with Mark = 8 to select the direction of normal cut, if BExtent is true.
7. [IFeatureManager::FinishSMNormalCut](#).
8. [ISMNormalCutFeatureData::Offset](#) to change the profile of the Normal Cut.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[ISMNormalCutFeatureData Interface](#)

Availability

SOLIDWORKS 2018 FCS, Revision Number 26.0

SOLIDWORKS API Help

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AddVariablePitchHelixFirstPitchAndDiameter Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : AddVariablePitchHelixFirstPitchAndDiameter Method (IFeatureManager)

Adds the first segment to a variable-pitch helix.

.NET Syntax

Visual Basic (Declaration)

```

Function AddVariablePitchHelixFirstPitchAndDiameter( _
    ByVal FirstPitch As System.Double, _
    ByVal FirstDiameter As System.Double _
) As System.Boolean

```

Visual Basic (Usage)

```

Dim instance As IFeatureManager
Dim FirstPitch As System.Double
Dim FirstDiameter As System.Double
Dim value As System.Boolean

value = instance.AddVariablePitchHelixFirstPitchAndDiameter(FirstPitch, FirstDiameter)

```

C#

```

System.bool AddVariablePitchHelixFirstPitchAndDiameter(
    System.double FirstPitch,
    System.double FirstDiameter
)

```

C++/CLI

```

System.bool AddVariablePitchHelixFirstPitchAndDiameter(
    & System.double FirstPitch,
    & System.double FirstDiameter
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FirstPitch

Pitch, which determines the width of one complete helix turn, measured parallel to the axis of the helix

FirstDiameter

Diameter, which determines how far the variable-pitch helix segment extends

Return Value

True if the first segment of the helix is added, false if not

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::AddVariablePitchHelixPatchAndDiameter](#).

Example

[Create Variable-pitch Helix \(C#\)](#)

[Create Variable-pitch Helix \(VB.NET\)](#)

[Create Variable-pitch Helix \(VBA\)](#)

[Create and Modify Variable-pitch Helix \(C#\)](#)

[Create and Modify Variable-pitch Helix \(VB.NET\)](#)

[Create and Modify Variable-pitch Helix \(VBA\)](#)

Remarks

To create and insert a variable-pitch helix, call:

1. [IFeatureManager::InsertVariablePitchHelix](#) to create it.
2. [IFeatureManager::AddVariablePitchHelixFirstPitchAndDiameter](#) to add the first segment.
3. [IFeatureManager::AddVariablePitchHelixSegment](#) one or more times to add the remaining segments.
4. [IFeatureManager::EndVariablePitchHelix](#) to insert it.

See Also

[IFeatureManager Interface](#)

[IFeatureManager Members](#)

[IHelixFeatureData Interface](#)

Availability

SOLIDWORKS 2011 FCS, Revision Number 19.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

AddVariablePitchHelixSegment Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : AddVariablePitchHelixSegment Method (IFeatureManager)

Adds a segment to a variable-pitch helix.

.NET Syntax

Visual Basic (Declaration)

```
Function AddVariablePitchHelixSegment( _  
    ByVal Height As System.Double, _  
    ByVal Diameter As System.Double, _  
    ByVal Pitch As System.Double _  
    ) As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As IFeatureManager  
Dim Height As System.Double  
Dim Diameter As System.Double  
Dim Pitch As System.Double  
Dim value As System.Boolean
```

```
value = instance.AddVariablePitchHelixSegment(Height, Diameter, Pitch)
```

C#



```
System.bool AddVariablePitchHelixSegment(  
    System.double Height,  
    System.double Diameter,  
    System.double Pitch  
)
```

C++/CLI

```
System.bool AddVariablePitchHelixSegment(  
& System.double Height,  
& System.double Diameter,  
& System.double Pitch  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Height

Helix segment revolution; 1.0 = 360 degrees

Diameter

Diameter, which determines how far the variable-pitch helix segment extends

Pitch

Pitch, which determines the width of one complete helix turn, measured parallel to the axis of the helix

Return Value

True if the variable-pitch helix segment is added, false if not

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::AddVariablePitchHelixSegment](#).

Example

[Create Variable-pitch Helix \(C#\)](#)

[Create Variable-pitch Helix \(VB.NET\)](#)

[Create Variable-pitch Helix \(VBA\)](#)

[Create and Modify Variable-pitch Helix \(C#\)](#)

[Create and Modify Variable-pitch Helix \(VB.NET\)](#)

[Create and Modify Variable-pitch Helix \(VBA\)](#)

Remarks

To create and insert a variable-pitch helix, call:

1. [IFeatureManager::InsertVariablePitchHelix](#) to create it.
2. [IFeatureManager::AddVariablePitchHelixFirstPitchAndDiameter](#) to add the first segment.
3. [IFeatureManager::AddVariablePitchHelixSegment](#) one or more times to add the remaining segments.
4. [IFeatureManager::EndVariablePitchHelix](#) to insert it.

See Also

[IFeatureManager Interface](#)

[IFeatureManager Members](#)

[IHelixFeatureData Interface](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

SOLIDWORKS API Help

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AdvancedHole Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : AdvancedHole Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager::AdvancedHole2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function AdvancedHole( _
    ByVal AdvancedHoleNearElementArray As System.Object, _
    ByVal AdvancedHoleFarElementArray As System.Object, _
    ByVal UseBaselineDimensions As System.Boolean, _
    ByVal IsCustomCallout As System.Boolean, _
    ByRef ResultArray As System.Object _
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager
Dim AdvancedHoleNearElementArray As System.Object
Dim AdvancedHoleFarElementArray As System.Object
Dim UseBaselineDimensions As System.Boolean
Dim IsCustomCallout As System.Boolean
Dim ResultArray As System.Object
Dim value As Feature

value = instance.AdvancedHole(AdvancedHoleNearElementArray, AdvancedHoleFarElementArray,
UseBaselineDimensions, IsCustomCallout, ResultArray)
```

C#

```
Feature AdvancedHole(
    System.object AdvancedHoleNearElementArray,
    System.object AdvancedHoleFarElementArray,
    System.bool UseBaselineDimensions,
    System.bool IsCustomCallout,
    out System.object ResultArray
)
```

C++/CLI

```
Feature^ AdvancedHole(
    & System.Object^ AdvancedHoleNearElementArray,
    & System.Object^ AdvancedHoleFarElementArray,
    & System.bool UseBaselineDimensions,
    & System.bool IsCustomCallout,
    [Out] System.Object^ ResultArray
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

AdvancedHoleNearElementArray
Array of near side hole data objects (see **Remarks**)

AdvancedHoleFarElementArray
Array of far side hole data objects (see **Remarks**)

UseBaselineDimensions
True to use baseline dimensions, false to not (see **Remarks**)

IsCustomCallout
True to customize the hole callout, false to use a default hole callout (see **Remarks**)

ResultArray
Array of result codes as defined in [swAdvancedHoleResults_e](#) (see **Remarks**)

Return Value

[IFeature](#): Nothing or null if this method results in an invalid geometry condition (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::AdvancedHole](#).

Remarks

The Advanced Hole feature:

- is separate from the Hole Wizard functionality (see [FeatureManager::HoleWizard5](#)).
- consists of several hole elements, e.g., counterbore, countersink, straight, and tapered tap, that are stacked from the near and far side faces inward. The near and far side stacks, combined, form the Advanced Hole.
- can be added only to parts at this time.

AdvancedHoleNearElementArray and AdvancedHoleFarElementArray each contain one or more hole type-specific objects in order as they stack from the near and far side faces:

- [ICounterboreElementData](#)

- [ICountersinkElementData](#)
- [IStraightElementData](#)
- [IStraightTapElementData](#)
- [ITaperedTapElementData](#)

The hole type-specific interfaces above extend the parent interface, [IAdvancedHoleElementData](#), which you use to set general Advanced Hole properties.

UseBaseLineDimensions controls which hole type-specific properties are valid. See [IAdvancedHoleFeatureData::UseBaselineDimensions](#).

IsCustomCallout set to true allows you to set a custom callout string for individual hole elements using [IAdvancedHoleElementData::CalloutString](#).

ResultArray contains a return code for each element in AdvancedHoleNearElementArray and AdvancedHoleFarElementArray. It is possible for this method to fail to create an advanced hole, even though ResultArray contains all success result codes. If the individual hole elements all combine to produce invalid geometry (e.g., a disjoint body or a self-intersecting face) or if incorrect near or far side faces are selected, then this method returns Nothing or null.

To create an Advanced Hole feature, follow the examples which do the following:

1. Call [IModelDocExtension::SelectByRay](#) to select the near side (Mark = 256) and far side (Mark = 512) faces of the Advanced Hole. You must select these faces by location, so do not use [IEntity::Select4](#).
2. Call [IModelDocExtension::CreateAdvancedHoleElementData](#) to create an AdvancedHoleElementData object.
3. Set the general Advanced Hole properties using the AdvancedHoleElementData object.
4. Cast the AdvancedHoleElementData object using one of the hole type-specific interfaces.
5. Set the specific Advanced Hole properties using the hole type-specific object.
6. Repeat steps 2 - 5 for each hole element you want to add to the near and far side stacks of the Advanced Hole.
7. Create near and far side arrays of the hole type-specific objects in stacking order from each face inward. The innermost elements of each stack should match to create a contiguous hole that extends from near side face to far side face.
8. Pass the near and far side arrays to IFeatureManager::AdvancedHole to create the Advanced Hole.
9. Delete the defining Advanced Hole sketch point and call [ISketchManager::CreatePoint](#) one or more times to insert the Advanced Hole in multiple locations.

To edit an Advanced Hole feature, call [IFeature::GetDefinition](#) to access [IAdvancedHoleFeatureData](#). See [Accessing Selections that Define Features](#).

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

Availability

SOLIDWORKS 2018 FCS, Revision Number 26.0

SOLIDWORKS API Help

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AdvancedHole2 Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : AdvancedHole2 Method (IFeatureManager)

Creates an Advanced Hole feature.

.NET Syntax

Visual Basic (Declaration)

```
Function AdvancedHole2( _
    ByVal AdvancedHoleNearElementArray As System.Object, _
    ByVal AdvancedHoleFarElementArray As System.Object, _
    ByVal UseBaselineDimensions As System.Boolean, _
    ByVal IsCustomCallout As System.Boolean, _
    ByVal IsDepthUptoTip As System.Boolean, _
    ByRef ResultArray As System.Object _
) As Feature
```

Visual Basic (Usage)

```
Dim instance As IFeatureManager
Dim AdvancedHoleNearElementArray As System.Object
Dim AdvancedHoleFarElementArray As System.Object
Dim UseBaselineDimensions As System.Boolean
Dim IsCustomCallout As System.Boolean
Dim IsDepthUptoTip As System.Boolean
Dim ResultArray As System.Object
Dim value As Feature
```

```
value = instance.AdvancedHole2(AdvancedHoleNearElementArray, AdvancedHoleFarElementArray,
    UseBaselineDimensions, IsCustomCallout, IsDepthUptoTip, ResultArray)
```

C#



```

Feature AdvancedHole2(
    System.object AdvancedHoleNearElementArray,
    System.object AdvancedHoleFarElementArray,
    System.bool UseBaselineDimensions,
    System.bool IsCustomCallout,
    System.bool IsDepthUptoTip,
    out System.object ResultArray
)

```

C++/CLI

```

Feature^ AdvancedHole2(
    & System.Object^ AdvancedHoleNearElementArray,
    & System.Object^ AdvancedHoleFarElementArray,
    & System.bool UseBaselineDimensions,
    & System.bool IsCustomCallout,
    & System.bool IsDepthUptoTip,
    & [Out] System.Object^ ResultArray
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

AdvancedHoleNearElementArray

Array of near side hole data objects (see **Remarks**)

AdvancedHoleFarElementArray

Array of far side hole data objects (see **Remarks**)

UseBaselineDimensions

True to use baseline dimensions, false to not (see **Remarks**)

IsCustomCallout

True to customize the hole callout, false to use a default hole callout (see **Remarks**)

IsDepthUptoTip

True if hole depth is up to the drill point tip, false if hole depth is up to the shoulder (full depth of hole) (see **Remarks**)

ResultArray

Array of result codes as defined in [swAdvancedHoleResults_e](#) (see **Remarks**)

Return Value

[IFeature](#): Nothing or null if this method results in an invalid geometry condition (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::AdvancedHole2](#).

Example

[Create Advanced Hole Feature \(VBA\)](#)

[Create Advanced Hole Feature \(VB.NET\)](#)

[Create Advanced Hole Feature \(C#\)](#)

Remarks

The Advanced Hole feature:

- is separate from the Hole Wizard functionality (see [FeatureManager::HoleWizard5](#)).
- consists of several hole elements, e.g., counterbore, countersink, straight, and tapered tap, that are stacked from the near and far side faces inward. The near and far side stacks, combined, form the Advanced Hole.
- can be added only to parts at this time.

[AdvancedHoleNearElementArray](#) and [AdvancedHoleFarElementArray](#) each contain one or more hole type-specific objects in order as they stack from the near and far side faces:

- [ICounterboreElementData](#)
- [ICountersinkElementData](#)
- [IStraightElementData](#)
- [IStraightTapElementData](#)
- [ITaperedTapElementData](#)

The hole type-specific interfaces above extend the parent interface, [IAdvancedHoleElementData](#), which you use to set general Advanced Hole properties.

[UseBaseLineDimensions](#) controls which hole type-specific properties are valid. See [IAdvancedHoleFeatureData::UseBaselineDimensions](#).

[IsCustomCallout](#) set to true allows you to set a custom callout string for individual hole elements using [IAdvancedHoleElementData::CalloutString](#).

[IsDepthUptoTip](#) is valid only when:

- No far side elements are in the stack.
- The near side end element:
 - is a hole or straight tap,
 - has an end shape of drill point, and
 - has an end condition that is blind or offset from surface.

ResultArray contains a return code for each element in AdvancedHoleNearElementArray and AdvancedHoleFarElementArray. It is possible for this method to fail to create an advanced hole, even though ResultArray contains all success result codes. If the individual hole elements all combine to produce invalid geometry (e.g., a disjoint body or a self-intersecting face) or if incorrect near or far side faces are selected, then this method returns Nothing or null.

To create an Advanced Hole feature, follow the examples which do the following:

1. Call [IModelDocExtension::SelectByRay](#) to select the near side (Mark = 256) and far side (Mark = 512) faces of the Advanced Hole. You must select these faces by location, so do not use [IEntity::Select4](#).
2. Call [IModelDocExtension::CreateAdvancedHoleElementData](#) to create an AdvancedHoleElementData object.
3. Set the general Advanced Hole properties using the AdvancedHoleElementData object.
4. Cast the AdvancedHoleElementData object using one of the hole type-specific interfaces.
5. Set the specific Advanced Hole properties using the hole type-specific object. For IStraightTapElementData and IStraightElementData objects, specify the IsDepthUptoTip requirements described in these Remarks.
6. Repeat steps 2 - 5 for each hole element you want to add to the near and far side stacks of the Advanced Hole.
7. Create near and far side arrays of the hole type-specific objects in stacking order from each face inward. The innermost elements of each stack should match to create a contiguous hole that extends from near side face to far side face. Create an empty far side array if using IsDepthUptoTip.
8. Pass the near and far side arrays to this method to create the Advanced Hole.
9. Delete the defining Advanced Hole sketch point and call [ISketchManager::CreatePoint](#) one or more times to insert the Advanced Hole in multiple locations.

To edit an Advanced Hole feature, call [IFeature::GetDefinition](#) to access [AdvancedHoleFeatureData](#). See [Accessing Selections that Define Features](#).

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

Availability

SOLIDWORKS 2020 FCS, Revision Number 28.0

SOLIDWORKS API Help

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ComponentPrimaryIdentifier Property (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : ComponentPrimaryIdentifier Property (IFeatureManager)

Gets the primary element displayed for the components in the FeatureManager design tree.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property ComponentPrimaryIdentifier As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager
Dim value As System.Integer

value = instance.ComponentPrimaryIdentifier
```

C#

```
System.int ComponentPrimaryIdentifier {get;}
```

C++/CLI

```
property System.int ComponentPrimaryIdentifier {
    System.int get();
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Primary identifier as defined by [swComponentIdentifier_e](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::ComponentPrimaryIdentifier](#).

Remarks

This property works in both SOLIDWORKS Design and [SOLIDWORKS Design connected to the 3DEXPERIENCE platform](#).

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IFeatureManager::SetComponentIdentifiers Method \(\)](#)

Availability

SOLIDWORKS 2022 SP01, Revision Number 30.1

SOLIDWORKS API Help

[Send comments on this topic.](#)

ComponentSecondaryIdentifier Property (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : ComponentSecondaryIdentifier Property (IFeatureManager)

Gets the secondary element(s) displayed for the components in the FeatureManager design tree.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property ComponentSecondaryIdentifier As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim value As System.Integer
```

```
value = instance.ComponentSecondaryIdentifier
```

C#

```
System.int ComponentSecondaryIdentifier {get;}
```

C++/CLI

```
property System.int ComponentSecondaryIdentifier {  
    System.int get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Secondary identifier(s) as defined by [swComponentIdentifier](#) e

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::ComponentSecondaryIdentifier](#).

Remarks

This property works in both SOLIDWORKS Design and [SOLIDWORKS Design connected to the 3DEXPERIENCE platform](#).

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IFeatureManager::SetComponentIdentifiers Method \(\)](#)

Availability

SOLIDWORKS 2022 SP01, Revision Number 30.1

SOLIDWORKS API Help

[Send comments on this topic.](#)

ComponentTertiaryIdentifier Property (IFeatureManager)

[See Also](#)



Gets the tertiary element(s) displayed for the components in the FeatureManager design tree.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property ComponentTertiaryIdentifier As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager
Dim value As System.Integer

value = instance.ComponentTertiaryIdentifier
```

C#

```
System.int ComponentTertiaryIdentifier {get;}
```

C++/CLI

```
property System.int ComponentTertiaryIdentifier {
    System.int get();
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Tertiary identifier(s) as defined by [swComponentIdentifier_e](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::ComponentTertiaryIdentifier](#).

Remarks

This property works in both SOLIDWORKS Design and [SOLIDWORKS Design connected to the 3DEXPERIENCE platform](#).

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IFeatureManager::SetComponentIdentifiers Method\(\)](#)

Availability

SOLIDWORKS 2022 SP01, Revision Number 30.1

SOLIDWORKS API Help

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CreateCoordinateSystem Method (IFeatureManager)

[See Also](#) [Example](#)



SolidWorks.Interop.sldworks Namespace > IFeatureManager Interface : CreateCoordinateSystem Method (IFeatureManager)

Creates a coordinate system feature using the specified entities.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateCoordinateSystem( _
    ByVal OriginPointEntity As System.Object, _
    ByVal XAxisEntities As System.Object, _
    ByVal YAxisEntities As System.Object, _
    ByVal ZAxisEntities As System.Object _
) As Feature
```

Visual Basic (Usage)

```
Dim instance As IFeatureManager
Dim OriginPointEntity As System.Object
Dim XAxisEntities As System.Object
Dim YAxisEntities As System.Object
Dim ZAxisEntities As System.Object
Dim value As Feature

value = instance.CreateCoordinateSystem(OriginPointEntity, XAxisEntities, YAxisEntities,
ZAxisEntities)
```

C#

```
Feature CreateCoordinateSystem(
    System.object OriginPointEntity,
    System.object XAxisEntities,
    System.object YAxisEntities,
    System.object ZAxisEntities
)
```

C++/CLI

```
Feature^ CreateCoordinateSystem(
    & System.Object^ OriginPointEntity,
    & System.Object^ XAxisEntities,
    & System.Object^ YAxisEntities,
    & System.Object^ ZAxisEntities
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

OriginPointEntity

Entity (vertex, point, midpoint, or the default point of origin on a part or assembly) for the coordinate system origin

XAxisEntities

Array of entities (vertex, point, or midpoint; linear edge or sketch line; non-linear edge or sketch entity; or planar face) for the x axis

YAxisEntities

Array of entities (vertex, point, or midpoint; linear edge or sketch line; non-linear edge or sketch entity; or planar face) for the y axis

ZAxisEntities

Array of entities (vertex, point, or midpoint; linear edge or sketch line; non-linear edge or sketch entity; or planar face) for the z axis

Return Value

[Feature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::CreateCoordinateSystem](#).

Example

[Create Coordinate System Feature \(VBA\)](#)

See Also

[IFeatureManager Interface](#)

[IFeatureManager Members](#)

[ICoordinateSystemFeatureData Interface](#)

[IFeatureManager::InsertCoordinateSystem Method](#)

[IFeatureManager::CreateCoordinateSystemUsingNumericalValues Method \(\)](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

CreateCoordinateSystemUsingNumericalValues Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : CreateCoordinateSystemUsingNumericalValues Method (IFeatureManager)

Creates a coordinate system feature using the specified numerical values for position and orientation with respect to the global coordinate system.

Visual Basic (Declaration)

```
Function CreateCoordinateSystemUsingNumericalValues( _
    ByVal UseLocation As System.Boolean, _
    ByVal DeltaX As System.Double, _
    ByVal DeltaY As System.Double, _
    ByVal DeltaZ As System.Double, _
    ByVal UseRotation As System.Boolean, _
    ByVal AngleX As System.Double, _
    ByVal AngleY As System.Double, _
    ByVal AngleZ As System.Double _
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager
Dim UseLocation As System.Boolean
Dim DeltaX As System.Double
Dim DeltaY As System.Double
Dim DeltaZ As System.Double
Dim UseRotation As System.Boolean
Dim AngleX As System.Double
Dim AngleY As System.Double
Dim AngleZ As System.Double
Dim value As Feature
```

```
value = instance.CreateCoordinateSystemUsingNumericalValues(UseLocation, DeltaX, DeltaY, DeltaZ,
UseRotation, AngleX, AngleY, AngleZ)
```

C#

```
Feature CreateCoordinateSystemUsingNumericalValues(
    System.bool UseLocation,
    System.double DeltaX,
    System.double DeltaY,
    System.double DeltaZ,
    System.bool UseRotation,
    System.double AngleX,
    System.double AngleY,
    System.double AngleZ
)
```

C++/CLI

```
Feature^ CreateCoordinateSystemUsingNumericalValues(
    & System.bool UseLocation,
    & System.double DeltaX,
    & System.double DeltaY,
    & System.double DeltaZ,
    & System.bool UseRotation,
    & System.double AngleX,
    & System.double AngleY,
    & System.double AngleZ
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

UseLocation

True to define the position of the coordinate system using numerical values, false to not

DeltaX

Distance in meters along the x-axis of the global coordinate system; valid only if UseLocation is true

DeltaY

Distance in meters along the y-axis of the global coordinate system; valid only if UseLocation is true

DeltaZ

Distance in meters along the z-axis of the global coordinate system; valid only if UseLocation is true

UseRotation

True to define the orientation of the coordinate system using numerical values, false to not

AngleX

0 <= Rotation in radians from the x-axis of the global coordinate system <= 6.283 radians; valid only if UseRotation is true

AngleY

0 <= Rotation in radians from the y-axis of the global coordinate system <= 6.283 radians; valid only if UseRotation is true

AngleZ

0 <= Rotation in radians from the z-axis of the global coordinate system <= 6.283 radians; valid only if UseRotation is true

Return Value

[IFeature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::CreateCoordinateSystemUsingNumericalValues](#).

Example

```
'VBA
'=====
'Preconditions:
'1. Open a new part.
'2. Open the Immediate window.
'Postconditions:
'1. Creates a new coordinate system at the specified position and angle relative to
the global coordinate system.
'2. Select Coordinate System1 in the FeatureManager design tree.
'3. Inspect the Immediate window and graphics area.
'=====
Option Explicit
Sub main()
    Dim swApp      As SldWorks.SldWorks
    Dim swModel    As SldWorks.ModelDoc2
    Dim swFeatMgr  As FeatureManager
    Dim swCoordFeat As SldWorks.Feature
    Dim posVal, angVal As Double
    Dim pos, ang   As Double
    Set swApp = Application.SldWorks
    Set swModel = swApp.ActiveDoc
    Set swFeatMgr = swModel.FeatureManager

    pos = 200
    posVal = pos / 1000 ' Position in meters

    ' 1 radian = 180°/p = 57.295779513° or approximately 57.3°
    ang = 60
    angVal = ang / 57.295779513 ' Angle in radians

    Debug.Print "Position(mm) : [" & pos & ", " & pos & ", " & pos & "]" Angle(deg) :
[" & ang & ", " & ang & ", " & ang & "]"

    Set swCoordFeat = swFeatMgr.CreateCoordinateSystemUsingNumericalValues(True,
posVal, posVal, posVal, True, angVal, angVal, angVal)
    Debug.Print "Name of the coordinate system feature : " & swCoordFeat.Name
```

End Sub

Remarks

This method is valid only for parts and assemblies.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IFeatureManager::CreateCoordinateSystem Method \(\)](#)
[IFeatureManager::InsertCoordinateSystem Method \(\)](#)
[ICoordinateSystemFeatureData Interface](#)

Availability

SOLIDWORKS 2022 FCS, Revision Number 30

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateCustomBendAllowance Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : CreateCustomBendAllowance Method (IFeatureManager)

Creates a custom bend allowance to use when creating a sheet metal feature.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateCustomBendAllowance() As CustomBendAllowance
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim value As CustomBendAllowance  
  
value = instance.CreateCustomBendAllowance()
```

C#

```
CustomBendAllowance CreateCustomBendAllowance()
```

C++/CLI

```
CustomBendAllowance^ CreateCustomBendAllowance();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Pointer to the [ICustomBendAllowance](#) object

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::CreateCustomBendAllowance](#).

Example

[Insert Sheet Metal Hem \(VBA\)](#)
[Insert Sheet Metal Hem \(VB.NET\)](#)
[Insert Sheet Metal Hem \(C#\)](#)

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[BendsFeatureData::SetCustomBendAllowance Method](#)
[IEdgeFlangeFeatureData::SetCustomBendAllowance Method](#)
[IFeature::InsertSheetMetalBaseFlange Method](#)
[IHemFeatureData::SetCustomBendAllowance Method](#)
[IJogFeatureData::SetCustomBendAllowance Method](#)
[IMiterFlangeFeatureData::SetCustomBendAllowance Method](#)
[IOneBendFeatureData::SetCustomBendAllowance Method](#)
[ISheetMetalFeatureData::SetCustomBendAllowance Method](#)
[ISketchBendFeatureData::SetCustomBendAllowance Method](#)
[IFeatureManager::InsertSheetMetalHem2 Method](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

CreateDefinition Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : CreateDefinition Method (IFeatureManager)

Creates a feature data object of the specified type.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateDefinition( _  
    ByVal Type As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim Type As System.Integer  
Dim value As System.Object  
  
value = instance.CreateDefinition(Type)
```

C#

```
System.object CreateDefinition(  
    System.int Type  
)
```

C++/CLI

```
System.Object^ CreateDefinition(  
& System.int Type  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Type

Feature name ID as defined in [swFeatureNameID_e](#):

- swFmBaseFlange (sheet metal base flange)
- swFmBeltAndChain (belt/chain)
- swFmBoundingBox (bounding box)
- swFmCirPattern (circular pattern)
- swFmCornerRelief (sheet metal corner relief)
- swFmCurvePattern (curve-driven pattern)
- swFmDerivedLPattern (derived-driven pattern)
- swFmDimPattern (variable/dimension pattern)
- swFmEdgeFlange (sheet metal edge flange)
- swFmFillet (constant radius, face, full round fillet/chamfer)
- swFmFillPattern (fill pattern)
- swFmFormToolInstance (library form tool)
- swFmGroundPlane (ground plane)
- swFmLibraryFeature (library)
- swFmLocalChainPattern (chain component pattern)
- swFmLocalCirPattern (circular component pattern)
- swFmLocalCurvePattern (curve-driven component pattern)
- swFmLocalLPattern (linear component pattern)
- swFmLocalSketchPattern (sketch-driven component pattern)
- swFmLPattern (linear pattern)

- swFmMateController (mate controller)
- swFmMirrorComponent (mirror components)
- swFmNormalCut (sheet metal normal cut)
- swFmRefCurve (projection curve)
- swFmRefSurface (surface sweep)
- swFmSketchBend (sheet metal sketched bend)
- swFmSketchPattern (sketch-driven pattern)
- swFmSMGusset (sheet metal gusset)
- swFmSolidToSheetMetal (convert solid to sheet metal)
- swFmSweep (boss sweep)
- swFmSweepCut (cut sweep)
- swFmSweepThread (sweep thread)
- swFmSweptFlange (sheet metal swept flange)
- swFmTabAndSlot (tab and slot)
- swFmTablePattern (table pattern)

Return Value

[thread](#), [sweep](#), [library](#), [library form tool](#), [tab and slot](#), [bounding box](#), [ground plane](#), [mirror components](#), [projection curve](#), [sheet metal normal cut](#), [sheet metal swept flange](#), [sheet metal gusset](#), [sheet metal edge flange](#), [convert solid to sheet metal](#), [simple fillet/chamfer](#), [belt/chain](#), [sheet metal base flange](#), [sheet metal corner relief](#), [sheet metal sketched bend](#), [mate controller](#), or pattern-specific feature data object (see **Remarks**); Nothing or null otherwise

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::CreateDefinition](#).

Example

See examples for:

[IBaseFlangeFeatureData](#)

[IBeltChainFeatureData](#)

[ICornerReliefFeatureData](#)

[IEdgeFlangeFeatureData](#)

[IMateControllerFeatureData](#)

[IMirrorComponentFeatureData](#)

[IPartialEdgeFilletData](#)

[ISimpleFilletFeatureData2](#)

[ISketchedBendFeatureData](#)

[ISMGussetFeatureData](#)

[ISweptFlangeFeatureData](#)

[ITabAndSlotFeatureData](#)

Example

[Create a Thread Feature \(VBA\)](#)
[Create a Thread Feature \(VB.NET\)](#)
[Create a Thread Feature \(C#\)](#)

Remarks

This method initializes the feature data objects with default data for pattern, sweep, bounding box, ground plane, mirror components, projection curve, sheet metal normal cut, sheet metal swept flange, sheet metal gusset, sheet metal edge flange, tab/slot, library form tool, and belt/chain features.

For sheet metal base flange, sheet metal corner relief, convert solid to sheet metal, library, simple fillet, and thread features, you must initialize feature data objects using specific initialize methods.

For mate controller features, you can either pre-select mates before calling this method or initialize the feature data object returned by this method with default values.

See the **See Also** section.

For additional information, see:

- [Library Features and LibraryFeatureData Objects](#)
- [Pattern Features and their Feature Data Objects](#)
- [Sweep Features and SweepFeatureData Objects](#)
- [Thread Features and ThreadFeatureData Objects](#)

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IFeature::GetDefinition Method](#)

[IFeature::IGetDefinition Method](#)
[IFeatureManager::CreateFeature Method](#)
[ILibraryFeatureData::Initialize Method](#)
[IThreadFeatureData::InitializeThreadData Method](#)
[ISimpleFilletFeatureData2::Initialize Method](#)
[IBaseFlangeFeatureData::Initialize Method \(\)](#)
[ICornerReliefFeatureData::Initialize](#)
[IMateControllerFeatureData::Initialize Method \(\)](#)
[IConvertSolidFeatureData::Initialize Method \(\)](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

SOLIDWORKS API Help

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CreateFeature Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : CreateFeature Method (IFeatureManager)

Creates the specified feature.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateFeature( _  
    ByVal FeatureData As System.Object _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim FeatureData As System.Object  
Dim value As Feature  
  
value = instance.CreateFeature(FeatureData)
```

C#

```
Feature CreateFeature(  
    System.object FeatureData  
)
```

C++/CLI

```
Feature^ CreateFeature(  
& System.Object^ FeatureData  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FeatureData

[thread](#), [sweep](#), [library](#), [library form tool](#) (see **Remarks**), [tab/slot](#), [bounding box](#), [ground plane](#), [mirror components](#), [projection curve](#), [sheet metal normal cut](#), [sheet metal swept flange](#), [sheet metal gusset](#), [sheet metal edge flange](#), [convert solid to sheet metal](#) (see **Remarks**), [simple fillet/chamfer](#), [belt/chain](#), [sheet metal base flange](#), [sheet metal corner relief](#), [sheet metal sketched bend](#), [mate controller](#), or a pattern-specific feature data object (see **Remarks**)

Return Value

[IFeature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::CreateFeature](#).

Example

See examples for:

[IBaseFlangeFeatureData](#)

[IBeltChainFeatureData](#)

[ICornerReliefFeatureData](#)

[IEdgeFlangeFeatureData](#)
[IMateControllerFeatureData](#)
[IMirrorComponentFeatureData](#)
[IPartialEdgeFilletData](#)
[ISimpleFilletFeatureData2](#)
[ISketchedBendFeatureData](#)
[ISMGussetFeatureData](#)
[ISweptFlangeFeatureData](#)
[ITabAndSlotFeatureData](#)

Example

[Create a Thread Feature \(VBA\)](#)
[Create a Thread Feature \(VB.NET\)](#)
[Create a Thread Feature \(C#\)](#)

Remarks

Before calling this method to create ConvertSolid and Library form tool features, ensure that the Sheet Metal tab is visible on the CommandManager toolbar.

Use:

- [IFeature::GetErrorCode2](#) to get any errors that occurred during the creation of the feature.
- [IFeatureManager::GetCreateFeatureErrors](#) to get any errors that occurred during the creation of a Mate Controller feature.

For additional information, see:

- [Library Features and LibraryFeatureData Objects](#)
- [Pattern Features and their Feature Data Objects](#)
- [Sweep Features and SweepFeatureData Objects](#)
- [Thread Features and ThreadFeatureData Objects](#)

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IFeature::GetDefinition Method](#)
[IFeatureManager::CreateDefinition Method](#)
[ILibraryFeatureData::Initialize Method](#)
[IThreadFeatureData::InitializeThreadData](#)
[ISimpleFilletFeatureData2::Initialize Method](#)
[IBaseFlangeFeatureData::Initialize Method \(\)](#)
[ICornerReliefFeatureData::Initialize Method \(\)](#)
[IMateControllerFeatureData::Initialize Method \(\)](#)
[IConvertSolidFeatureData::Initialize Method \(\)](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateFormTool Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : CreateFormTool Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager::CreateFormTool2](#).

.NET Syntax

Visual Basic (Declaration)

Function CreateFormTool() As Feature

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim value As Feature
```

```
value = instance.CreateFormTool()
```

C#

```
Feature CreateFormTool()
```

C++/CLI

```
Feature^ CreateFormTool();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Pointer to the [IFeature](#) object

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::CreateFormTool](#).

Remarks

To insert a forming tool from the Design Library, call [IFeatureManager::InsertFormToolFeature](#).

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateFormTool2 Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : CreateFormTool2 Method (IFeatureManager)

Creates a forming tool feature with the specified point of insertion in a sheet metal part.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateFormTool2( _  
    ByVal OriginX As System.Double, _  
    ByVal OriginY As System.Double _  
    ) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim OriginX As System.Double  
Dim OriginY As System.Double  
Dim value As Feature
```

```
value = instance.CreateFormTool2(OriginX, OriginY)
```

C#

```
Feature CreateFormTool2(  
    System.double OriginX,  
    System.double OriginY  
)
```

C++/CLI

```
Feature^ CreateFormTool2(  
&    System.double OriginX,
```

```
& System.double OriginY  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

OriginX
x coordinate of insertion point
OriginY
y coordinate of insertion point

Return Value

[IFeature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::CreateFormTool2](#).

Example

[Create Forming Tool \(VBA\)](#)
[Create Forming Tool \(VB.NET\)](#)
[Create Forming Tool \(C#\)](#)

Remarks

Before calling this method, call [IModelDocExtension::SelectByID2](#):

1. with Append = True and Mark = 1 to select the stopping face.
2. (optional) one or more times with Append = True and Mark = 2 to select the faces to remove.

To insert a forming tool from the Design Library, call [IFeatureManager::InsertFormToolFeature](#).

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

Availability

SOLIDWORKS 2013 FCS, Revision Number 21.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateSaveBodyFeature Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : CreateSaveBodyFeature Method (IFeatureManager)

Creates a Save Bodies feature and creates part and assembly documents of the save bodies.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateSaveBodyFeature( _  
    ByVal Bodies As System.Object, _  
    ByVal FilePaths As System.Object, _  
    ByVal AssemName As System.String, _  
    ByVal ConsumeBody As System.Boolean, _  
    ByVal CopyCustomProperty As System.Boolean _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim Bodies As System.Object  
Dim FilePaths As System.Object  
Dim AssemName As System.String  
Dim ConsumeBody As System.Boolean  
Dim CopyCustomProperty As System.Boolean
```

```
Dim value As System.Object
```

```
value = instance.CreateSaveBodyFeature(Bodies, FilePaths, AssemName, ConsumeBody,  
CopyCustomProperty)
```

C#

```
System.Object CreateSaveBodyFeature(  
    System.Object Bodies,  
    System.Object FilePaths,  
    System.String AssemName,  
    System.Bool ConsumeBody,  
    System.Bool CopyCustomProperty  
)
```

C++/CLI

```
System.Object^ CreateSaveBodyFeature(  
& System.Object^ Bodies,  
& System.Object^ FilePaths,  
& System.String^ AssemName,  
& System.Bool ConsumeBody,  
& System.Bool CopyCustomProperty  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Bodies

Array of [solid bodies](#) to save as parts (See **Remarks**)

FilePaths

Array of paths and filenames of the part documents to which to save Bodies

AssemName

Path and filename of the assembly document to which to save Bodies

ConsumeBody

See **Remarks**

CopyCustomProperty

See **Remarks**

Return Value

[Feature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::CreateSaveBodyFeature](#).

Example

[Create Save Bodies Feature and Create an Assembly \(VBA\)](#)

Remarks

If any solid bodies in Bodies are invalid, they are skipped/ignored.

For parameter...	Specify...
ConsumeBody	VARIANT_TRUE (-1) to consume all bodies in the original part, VARIANT_FALSE (0) to not
CopyCustomProperty	VARIANT_TRUE (-1) to copy custom properties to the new parts, VARIANT_FALSE (0) to not

See Also

[IFeatureManager Interface](#)

[IFeatureManager Members](#)

[ISaveBodyFeatureData Interface](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : CreateStructuralMemberGroup Method (IFeatureManager)

Creates a weldment structural-member group.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateStructuralMemberGroup() As StructuralMemberGroup
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager
Dim value As StructuralMemberGroup

value = instance.CreateStructuralMemberGroup()
```

C#

```
StructuralMemberGroup CreateStructuralMemberGroup()
```

C++/CLI

```
StructuralMemberGroup^ CreateStructuralMemberGroup();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[Structural-member group](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::CreateStructuralMemberGroup](#).

Example

[Insert Structural Weldment \(C#\)](#)
[Insert Structural Weldment \(VB.NET\)](#)
[Insert Structural Weldment \(VBA\)](#)
[Create Structural-Member Group \(VBA\)](#)
[Create Structural-Member Group \(VB.NET\)](#)
[Create Structural-Member Group \(C#\)](#)

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IStructuralMemberFeatureData::ISetGroups Method](#)
[IFeatureManager::InsertStructuralWeldment3 Method](#)

Availability

SOLIDWORKS 2009 SP5, Revision 17.5

SOLIDWORKS API Help

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CutListFolderLocation Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : CutListFolderLocation Method (IFeatureManager)

Gets the parent folder of the specified solid body / cut list item / body folder.

.NET Syntax

Visual Basic (Declaration)

```
Function CutListFolderLocation( _  
    ByVal QueryObj As System.Object _  
) As System.Object
```

```
Dim instance As IFeatureManager  
Dim QueryObj As System.Object  
Dim value As System.Object  
  
value = instance.CutListFolderLocation(QueryObj)
```

C#

```
System.object CutListFolderLocation(  
    System.object QueryObj.  
)
```

C++/CLI

```
System.Object^ CutListFolderLocation(  
    & System.Object^ QueryObj.  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

QueryObj

Solid Body / cut list item / body folder / sub-weldment folder [feature](#)

Return Value

Parent folder feature of QueryObj; null if QueryObj is not a Solid Body / cut list / body folder / sub-weldment folder

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::CutListFolderLocation](#).

Example

See the [IModelDocExtension::ReorderFeature2](#) examples.

Remarks

This method only works for:

- Solid Bodies / cut list items
- Feature / body folders
- Sub-weldment folders

Use [IFeatureManager::FeatureFolderLocation](#) to locate other FeatureManager design tree objects.

See Also

[IFeatureManager Interface](#)

[IFeatureManager Members](#)

[IModelDocExtension::ReorderFeature2 Method \(\)](#)

Availability

SOLIDWORKS 2025 SP02, Revision Number 33.2

SOLIDWORKS API Help

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Document Property (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : Document Property (IFeatureManager)

Gets the specified document.

.NET Syntax

Visual Basic (Declaration)

ReadOnly Property Document As ModelDoc2

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim value As ModelDoc2
```

```
value = instance.Document
```

C#

```
ModelDoc2 Document {get;}
```

C++/CLI

```
property ModelDoc2^ Document {  
    ModelDoc2^ get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

[Document](#) to get

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::Document](#).

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IModelDocExtension::Document Property](#)
[IModelViewManager::Document Property](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

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DraftXpertChange Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : DraftXpertChange Method (IFeatureManager)

Changes the parameters on the selected drafted faces, regardless of whether the drafted faces were created manually or with DraftXpert, provided that DraftXpert can process them.

.NET Syntax

Visual Basic (Declaration)

```
Function DraftXpertChange( _  
    ByVal Angle As System.Double, _  
    ByVal FlipDir As System.Boolean, _  
    ByVal EdgeDraft As System.Boolean, _  
    ByVal PropType As System.Integer, _  
    ByVal IsStepDraft As System.Boolean, _  
    ByVal IsBodyDraft As System.Boolean _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim Angle As System.Double  
Dim FlipDir As System.Boolean  
Dim EdgeDraft As System.Boolean  
Dim PropType As System.Integer  
Dim IsStepDraft As System.Boolean
```

```
Dim IsBodyDraft As System.Boolean
Dim value As Feature
```

```
value = instance.DraftXpertChange(Angle, FlipDir, EdgeDraft, PropType, IsStepDraft, IsBodyDraft)
```

C#

```
Feature DraftXpertChange(
    System.double Angle,
    System.bool FlipDir,
    System.bool EdgeDraft,
    System.int PropType,
    System.bool IsStepDraft,
    System.bool IsBodyDraft
)
```

C++/CLI

```
Feature^ DraftXpertChange(
    & System.double Angle,
    & System.bool FlipDir,
    & System.bool EdgeDraft,
    & System.int PropType,
    & System.bool IsStepDraft,
    & System.bool IsBodyDraft
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Angle

Draft angle in radians

FlipDir

True to flip the direction of the draft (the draft direction is determined by the selected direction, which could be a planar face, a reference plane, an edge, or two vertices; if you refer to the last argument in [IModelDocExtension::SelectByID2](#), then these items have a selection mark of 1; the faces to draft have a selection mark of 2, and the draft edges have a selection mark of 4), false to not

EdgeDraft

True if this is an edge draft and the user selects edges to draft, false if this is a face draft and the user selects faces to draft (SOLIDWORKS automatically selects the faces next to the edge to be drafted)

PropType

Propagation type:

- 0 = no propagation
- 1 = propagate to the next tangent faces that share the base neutral plane; valid for edge draft or neutral plane draft
- 2 = propagate to all faces that are a neighbor of the neutral plane
- 3 = propagate to all faces that are a neighbor of the neutral plane on all inner loops (for example, all faces of a pocket or a boss)
- 4 = propagate to all faces that are a neighbor of the neutral plane on the outer loop (for example, the sides of a box with the bottom face as the neutral plane)

IsStepDraft

True if the draft is a step draft, false if not

IsBodyDraft

True if the draft is a body draft, false if not

Return Value

Pointer to the modified draft [feature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::DraftXpertChange](#).

See Also

[IFeatureManager Interface](#)

[IFeatureManager Members](#)

[IFeatureManager::DraftXpertRemove Method](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

DraftXpertRemove Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : DraftXpertRemove Method (IFeatureManager)

Deletes the draft on the selected faces. If all the faces of a draft are selected, then this method deletes the draft feature; if not, then this method edits the draft feature and removes the selected face references from it.

.NET Syntax

Visual Basic (Declaration)

```
Function DraftXpertRemove() As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim value As Feature  
  
value = instance.DraftXpertRemove()
```

C#

```
Feature DraftXpertRemove()
```

C++/CLI

```
Feature^ DraftXpertRemove();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Pointer to the [feature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::DraftXpertRemove](#).

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IFeatureManager::DraftXpertChange Method](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

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EditDeleteFace Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : EditDeleteFace Method (IFeatureManager)

Edits a DeleteFace feature.

.NET Syntax

Visual Basic (Declaration)

```
Function EditDeleteFace(  
    ByVal Refill As System.Integer _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim Refill As System.Integer  
Dim value As System.Boolean
```

```
value = instance.EditDeleteFace(Refill)
```

C#

```
System.bool EditDeleteFace(  
    System.int Refill  
)
```

C++/CLI

```
System.bool EditDeleteFace(  
& System.int Refill  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Refill

- 1 refills the face after it is deleted
- 0 does not

Return Value

True if the operation succeeds, false if not

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::EditDeleteFace](#).

Remarks

This method assumes that the new faces to delete and the [DeleteFace](#) feature are selected.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IDeleteFaceFeatureData Interface](#)
[IModelDocExtension::InsertDeleteFace](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0